

Length & Time

Question Paper 1

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Length & Time
Booklet	Question Paper 1

Time allowed: 20 minutes

Score: /16

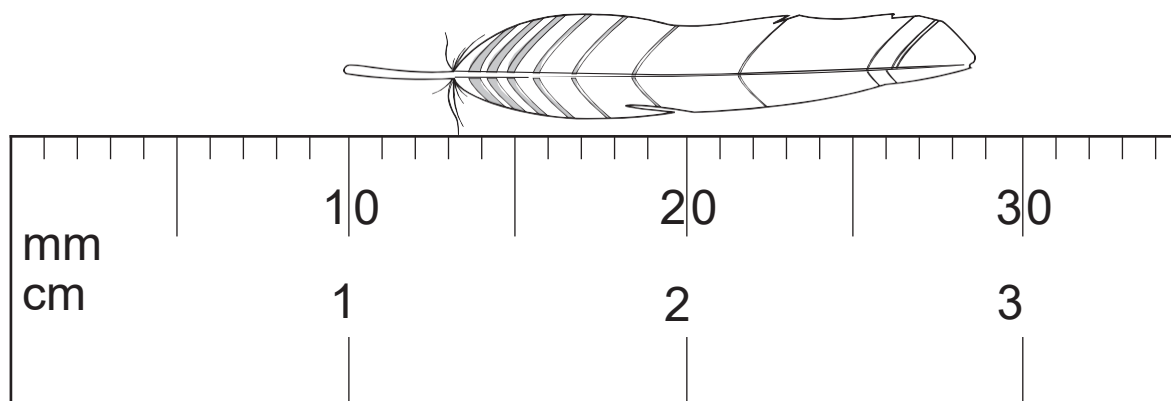
Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

The diagram shows an enlarged drawing of the end of a metre rule. It is being used to measure the length of a small feather.



What is the length of the feather?

- A 19 mm B 29 mm C 19 cm D 29 cm

Question 2

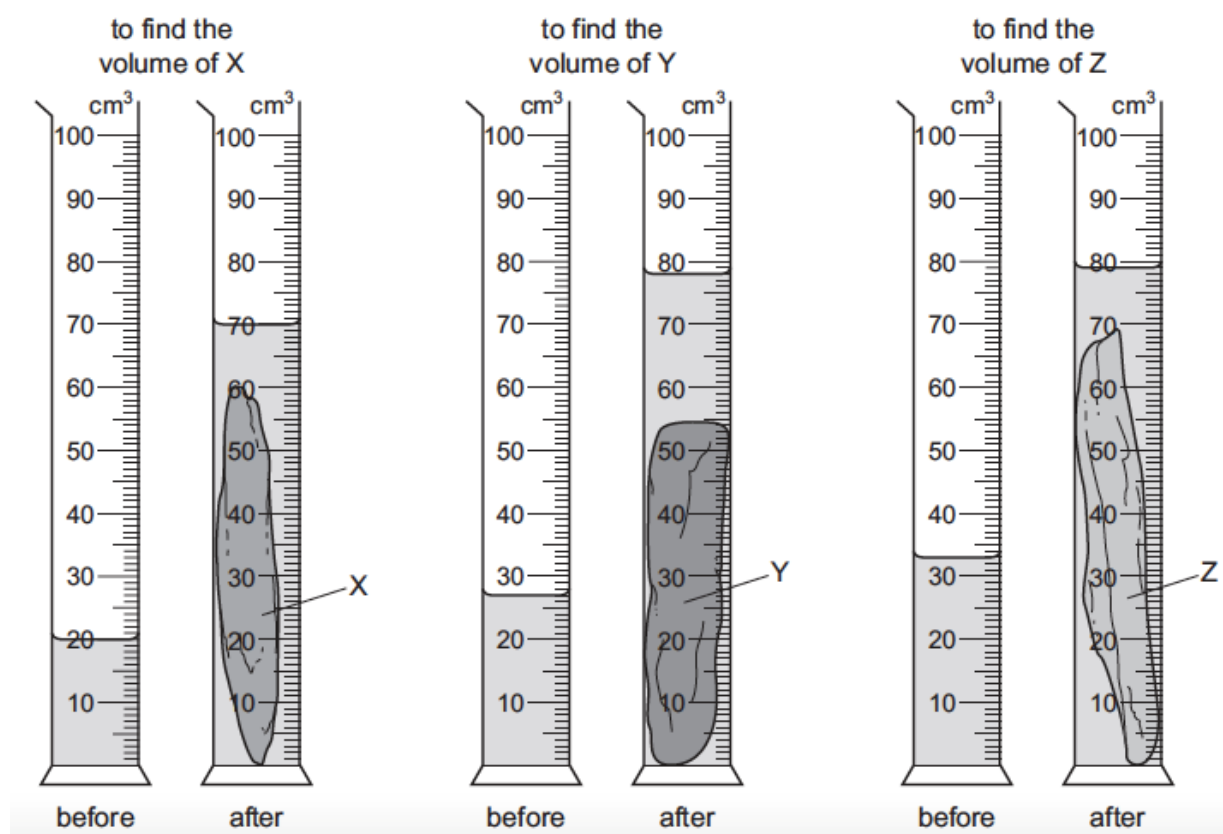
Which measurement can be made using a micrometer screw gauge?

- A. the air pressure of a tyre
- B. the diameter of a wire
- C. the turning effect of a spanner
- D. the wavelength of microwaves

Question 3

A geologist compares the volumes of three rocks, X, Y and Z. Three measuring cylinders contain different volumes of water. He places each rock into one of the measuring cylinders.

The diagrams show the measuring cylinders before and after the rocks are put in.

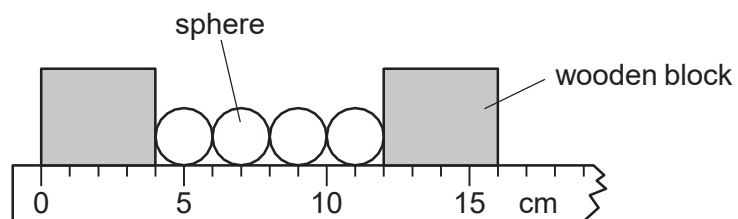


Which row shows the volumes of X, Y and Z in order, from largest to smallest?

	largest volume	→	smallest volume
A	X	Z	Y
B	Y	X	Z
C	Y	Z	X
D	Z	Y	X

Question 4

The diagram shows four identical spheres placed between two wooden blocks on a ruler.



What is the diameter of one sphere?

- A 1.0cm B 2.0cm C 3.0cm D 4.0cm

Question 5

A cook wants to prepare some food to be cooked by 1.15 p.m. He uses an oven with an automatic timer that can be set to switch on and off at certain times. The oven needs to be switched on for 2 hours 10 minutes.

At which time does the oven need to switch on?

- A 11.05 a.m. B 11.25 a.m. C 3.05 p.m. D 3.25 p.m.

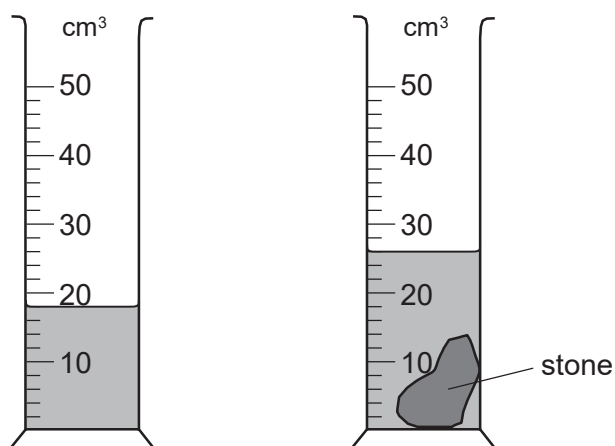
Question 6

Which option contains **only** apparatus that could be used to determine the volume of a small block of unknown material?

- A measuring cylinder, metre rule
- B measuring cylinder, stopwatch
- C metre rule, balance
- D metre rule, stopwatch

Question 7

The diagram shows a measuring cylinder used to measure the volume of a small stone.



What is the volume of the stone?

- A 8 cm³ B 9 cm³ C 14 cm³ D 26 cm³

Question 8

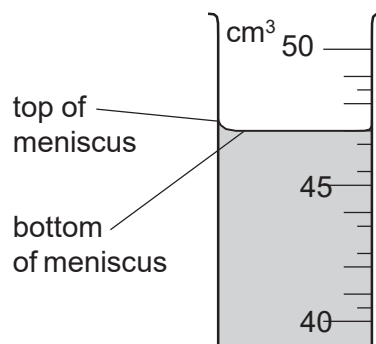
A student uses a measuring cylinder to measure the volume of a quantity of water.

Which action would make her result **less** accurate?

- A. making sure her eye is level with the water surface
- B. making sure the cylinder is vertical
- C. reading the bottom of the meniscus
- D. using the largest measuring cylinder possible

Question 9

A student uses a measuring cylinder to measure the volume of some water. The diagram shows part of the measuring cylinder. The top and bottom of the meniscus are labelled.

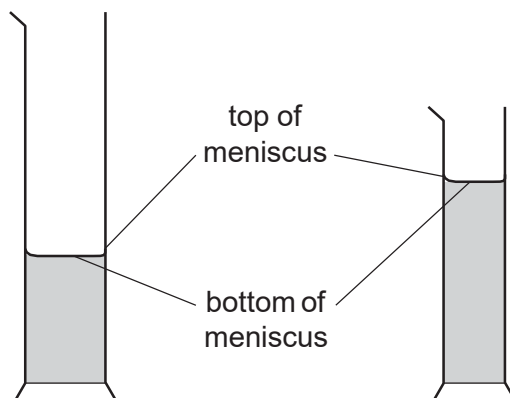


What is the volume of the water?

- A 47.0 cm^3 B 47.5 cm^3 C 49.0 cm^3 D 49.5 cm^3

Question 10

A student wishes to measure accurately the volume of approximately 40 cm^3 of water. She has two measuring cylinders, a larger one that can hold 100 cm^3 , and a smaller one that can hold 50 cm^3 . The water forms a meniscus where it touches the glass.



Which cylinder should the student use and which water level should she use to ensure an accurate result?

	cylinder	water level
A	larger one	bottom of meniscus
B	larger one	top of meniscus
C	smaller one	bottom of meniscus
D	smaller one	top of meniscus

Question 11

Diagram 1 shows a measuring cylinder containing water.

Five identical steel balls are now lowered into the measuring cylinder. Diagram 2 shows the new water level in the cylinder.

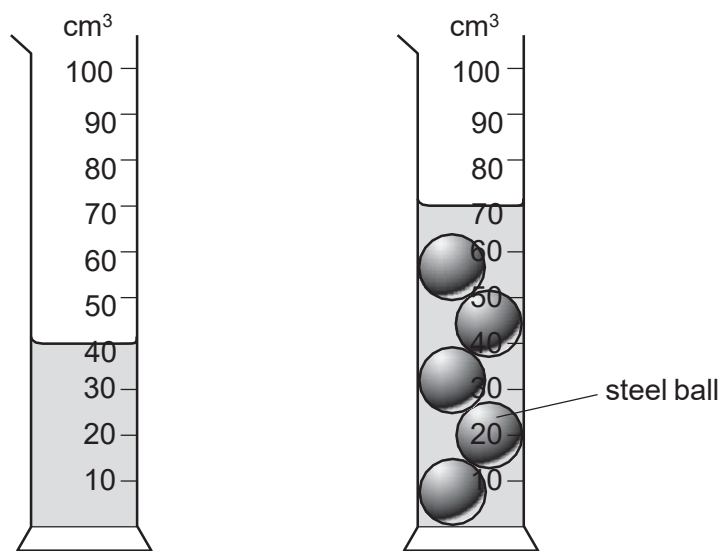


diagram 1

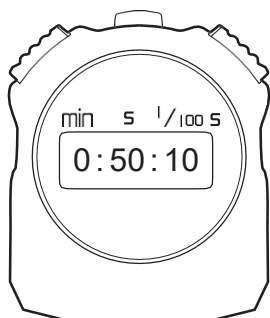
diagram 2

What is the volume of each steel ball?

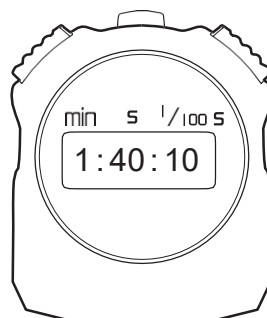
- A 6 cm³ B 14 cm³ C 30 cm³ D 70 cm³

Question 12

A stopwatch is used to time a runner in a race. The diagrams show the stopwatch at the start and at the end of a lap of the race.



start of lap



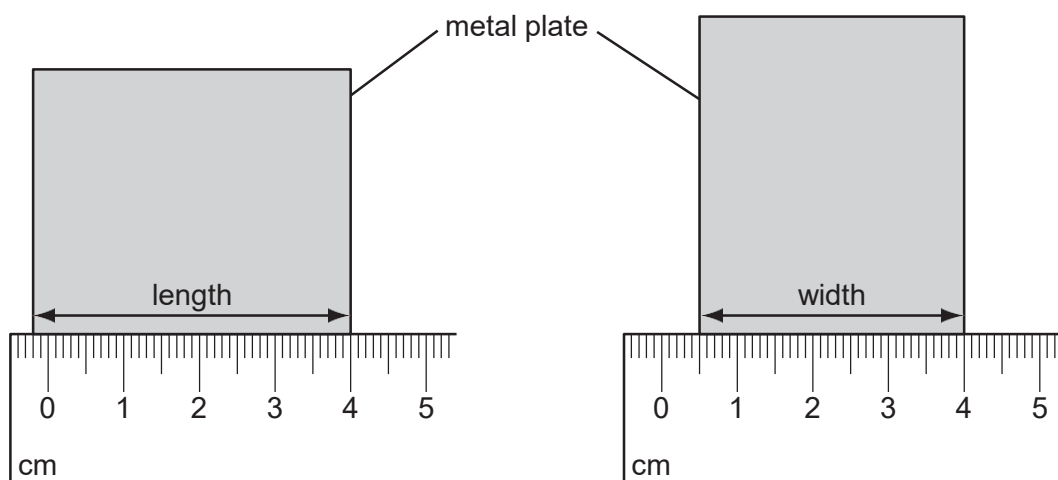
end of lap

How long did the runner take to finish the lap of the race?

- A. 50.00 seconds
- B. 50.10 seconds
- C. 90.00 seconds
- D. 100.10 seconds

Question 13

A student uses a ruler to measure the length and the width of a small rectangular metal plate.

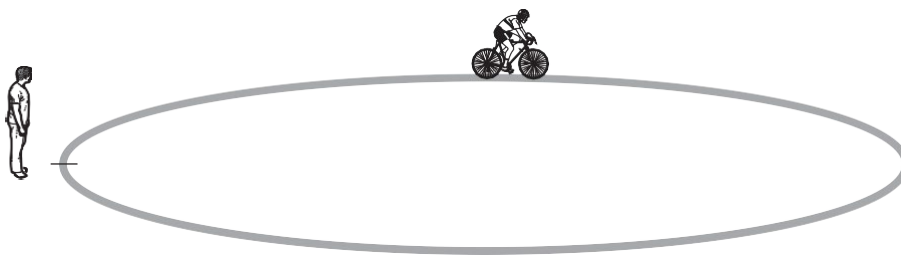


What is the area of the plate?

- A 14.0cm^2 B 14.7cm^2 C 16.0cm^2 D 16.8cm^2

Question 14

A cyclist rides round a track three times.



Her friend uses a stopwatch to record the time at the start of the ride, after one circuit, and at the end of the three circuits. The readings from the stopwatch are shown.



at the
start



after
one circuit



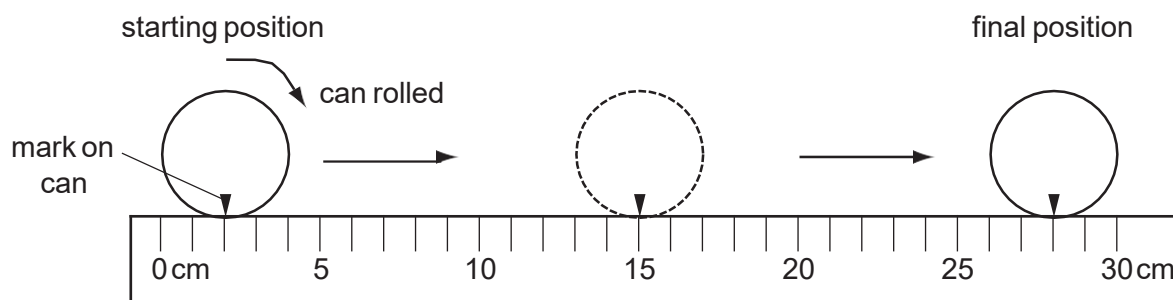
at the end of
three circuits

What is the average time for one circuit of the track?

- A 174 s B 180 s C 198 s D 200 s

Question 15

A cylindrical can is rolled along the ruler shown in the diagram.



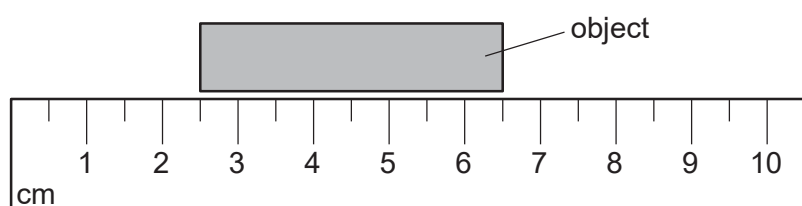
The can rolls over twice.

What is the circumference (distance all round) of the can?

- A 13 cm B 14 cm C 26 cm D 28 cm

Question 16

A ruler is used to measure the length of an object.



What is the length of the object?

- A 3.0cm B 4.0cm C 5.0cm D 6.5cm

Motion

Question Paper 1

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Motion
Booklet	Question Paper 1

Time allowed: 19 minutes

Score: /15

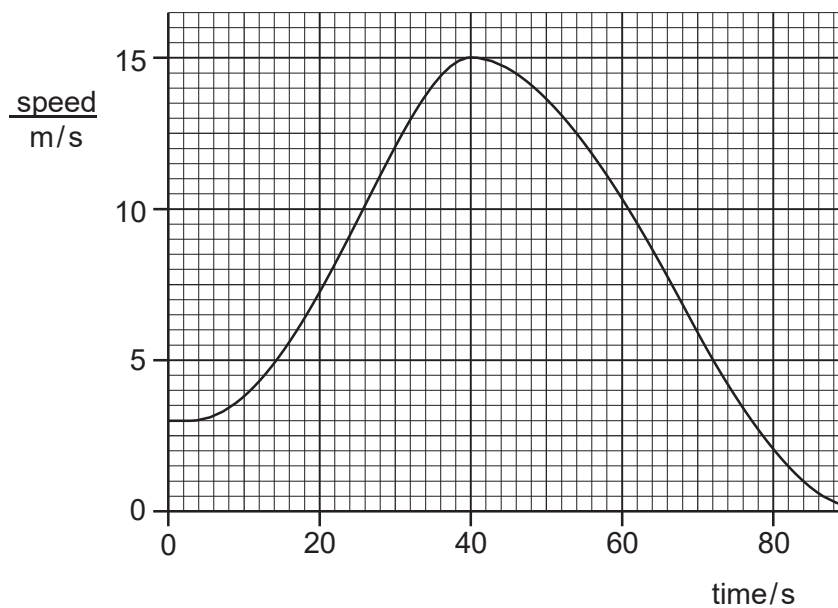
Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

The speed-time graph shown is for a car moving in a straight line.



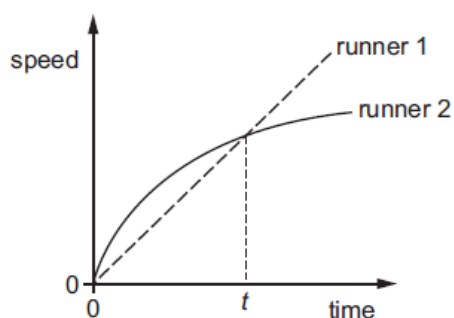
What is the acceleration of the car when the time is 40 s?

- A 0 m/s^2 B $\frac{15-3}{40} \text{ m/s}^2$ C $\frac{15}{40} \text{ m/s}^2$ D $(15-3) \text{ m/s}^2$

Question 2

Two runners take part in a race.

The graph shows how the speed of each runner changes with time.



What does the graph show about the runners at time t ?

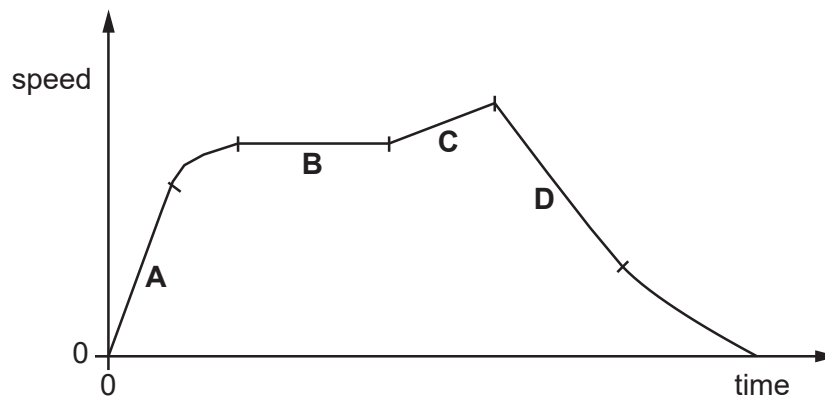
- A. Both runners are moving at the same speed.
- B. Runner 1 has zero acceleration.
- C. Runner 1 is overtaking runner 2.
- D. Runner 2 is slowing down.

Question 3

A car travels along a straight road.

The speed-time graph for this journey is shown.

During which labelled part of the journey is the resultant force on the car zero?



Question 4

A large stone is dropped from a bridge into a river. Air resistance can be ignored.

Which row describes the acceleration and the speed of the stone as it falls?

	acceleration of the stone	speed of the stone
A	constant	constant
B	constant	increasing
C	increasing	constant
D	increasing	increasing

Question 5

Below are four statements about acceleration.

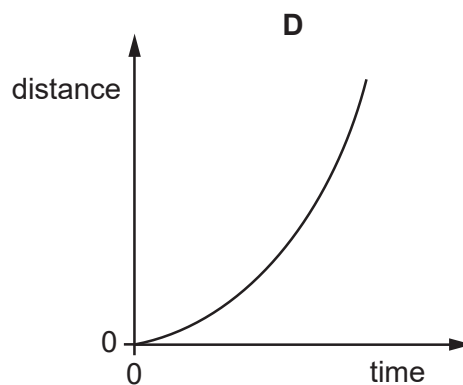
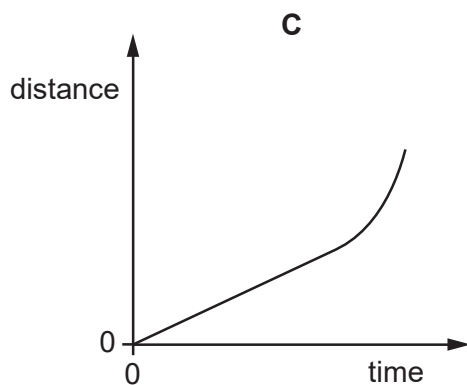
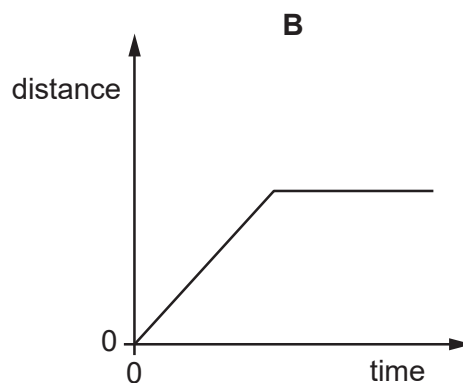
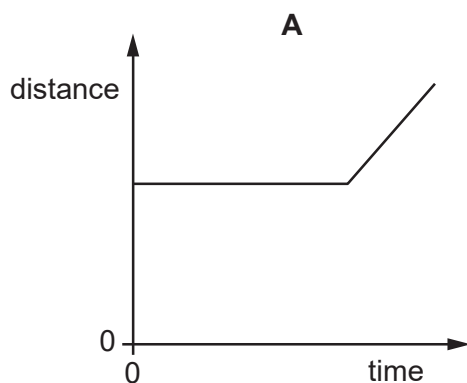
Which statement is not correct?

- A. Acceleration always involves changing speed.
- B. Changing direction always involves acceleration.
- C. Changing speed always involves acceleration.
- D. Circular motion always involves acceleration.

Question 6

An object moves at a constant speed for some time, then begins to accelerate.

Which distance-time graph shows this motion?



Question 7

A heavy object is released near the surface of the Earth and falls freely. Air resistance can be ignored.

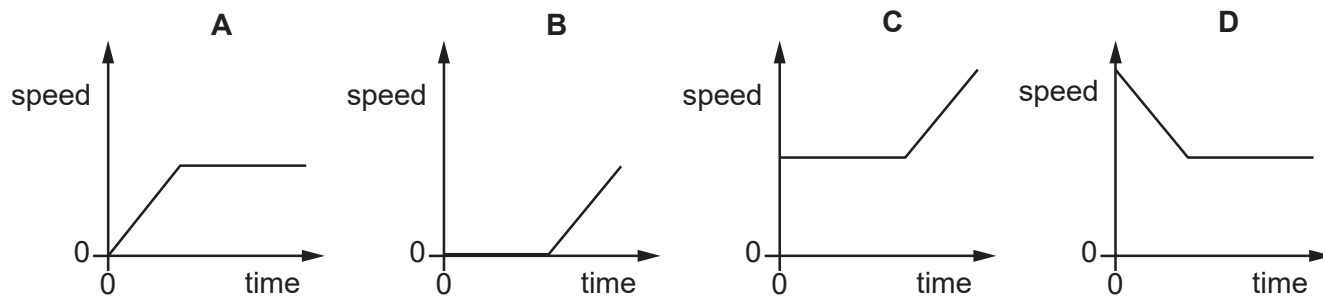
Which statement about the acceleration of the object due to gravity is correct?

- A The acceleration depends on the mass of the object.
- B The acceleration depends on the volume of the object.
- C The acceleration is constant.
- D The acceleration is initially zero and increases as the object falls.

Question 8

A car moves with constant speed and then constant acceleration.

Which graph is the speed-time graph for the car?



Question 9

A car travels 100 km. The journey takes two hours. The highest speed of the car is 80 km / h, and the lowest speed is 40 km/h.

What is the average speed for the journey?

- A 40 km/h B 50 km/h C 60 km/h D 120 km/h

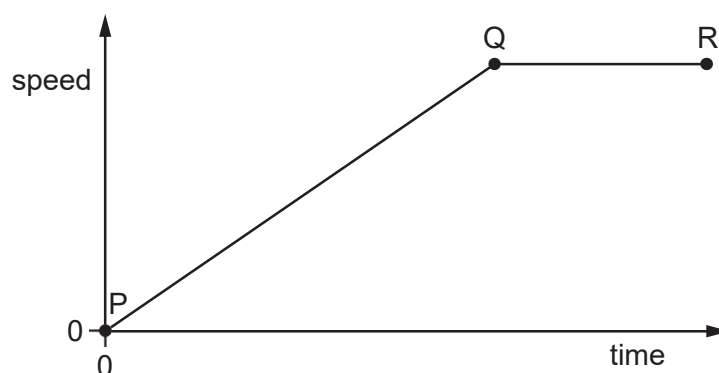
Question 10

What does the area under a speed-time graph represent?

- A. acceleration
- B. average speed
- C. deceleration
- D. distance travelled

Question 11

The speed-time graph shows the motion of a car.

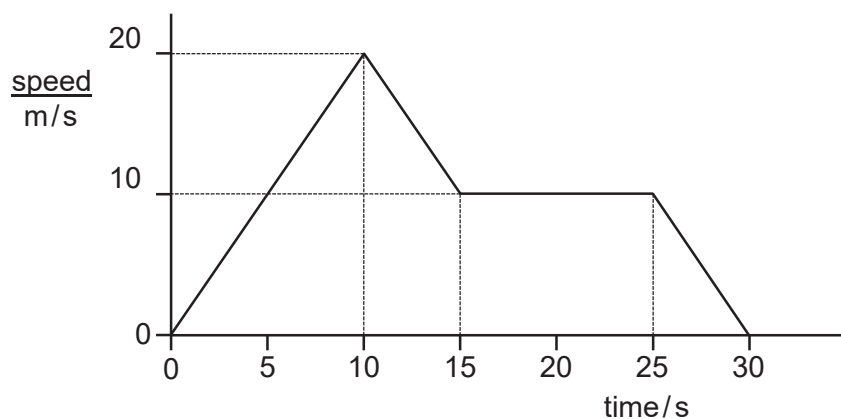


Which row describes the motion?

	between P and Q	between Q and R
A	accelerating	moving at constant speed
B	accelerating	not moving
C	moving at constant speed	decelerating
D	moving at constant speed	not moving

Question 12

The graph represents the motion of a car.

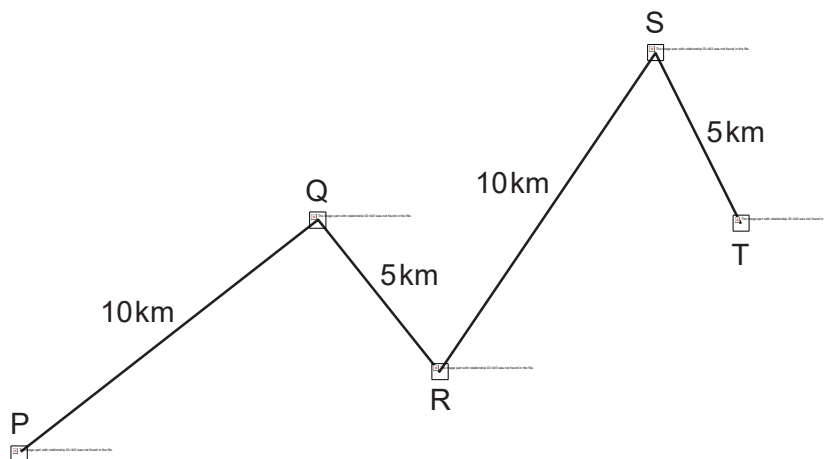


What is the distance travelled by the car while it is moving at a constant speed?

- A 100 m B 150 m C 250 m D 300 m

Question 13

A car travels along the route PQRST in 30 minutes.



What is the average speed of the car?

- A 10 km/hour B 20 km/hour C 30 km/hour D 60 km/hour

Question 14

The table shows the readings on a car speedometer at 5 second intervals.

time/s	speed km/h
0	0
5	30
10	50
15	60
20	65

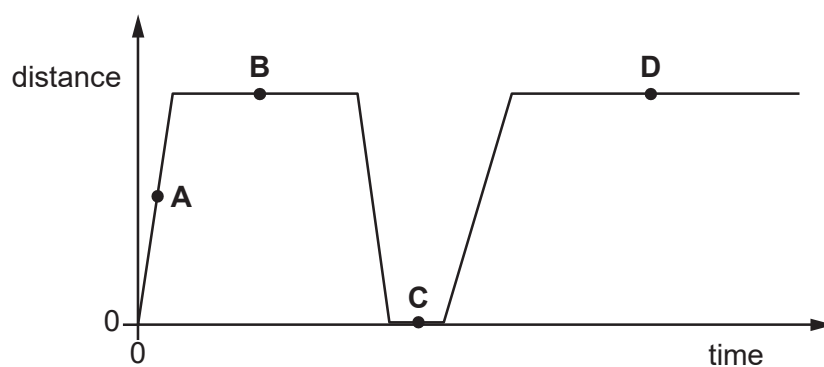
Which row describes the speed and the acceleration of the car?

	speed	acceleration
A	decreasing	zero
B	decreasing	not zero
C	increasing	zero
D	increasing	not zero

Question 15

The diagram shows the distance-time graph for a car.

At which labelled point is the car moving with constant speed?



Motion

Question Paper 2

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Motion
Booklet	Question Paper 2

Time allowed: 20 minutes

Score: /16

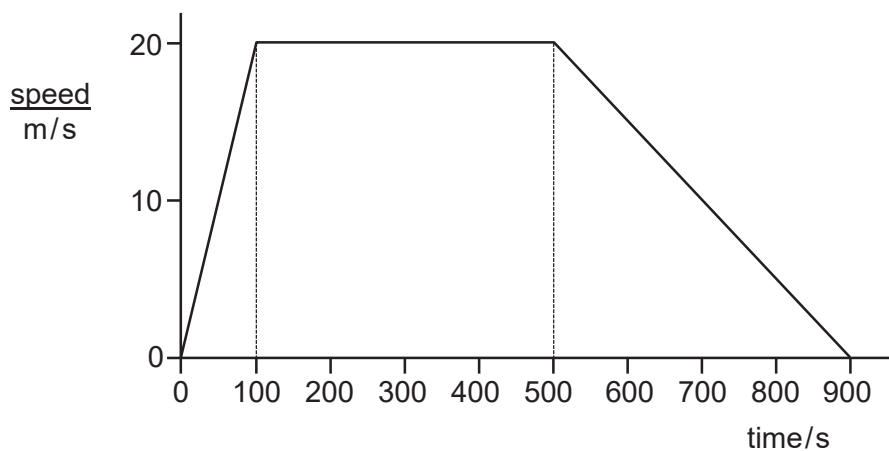
Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

The graph represents the motion of a train travelling between two stations.

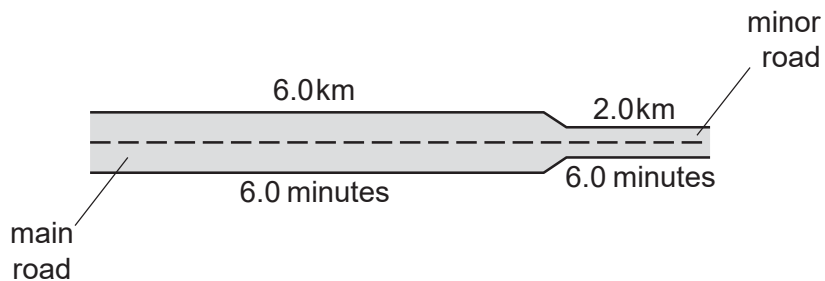


Which statement about the train is correct?

- A. Its acceleration takes a longer time than its deceleration.
- B. It travels at constant speed for less than half of its journey time.
- C. It travels 2000 m in the first 100 s.
- D. It travels 10 000 m at constant speed.

Question 2

A car travels 6.0 km along a main road in 6.0 minutes. It then travels 2.0 km along a minor road in 6.0 minutes.

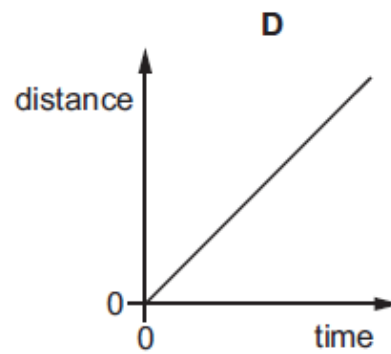
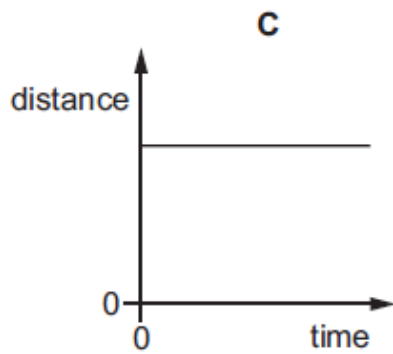
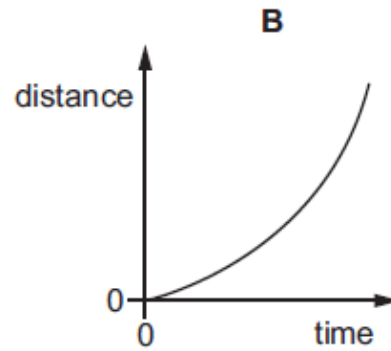


Which calculation of average speed for the whole journey is correct?

- A $8.0 \div 12.0 = 0.67 \text{ km/minute}$
- B $12.0 \div 8.0 = 1.5 \text{ km/minute}$
- C $8.0 + 12.0 = 20 \text{ km/minute}$
- D $8.0 \times 12.0 = 96 \text{ km/minute}$

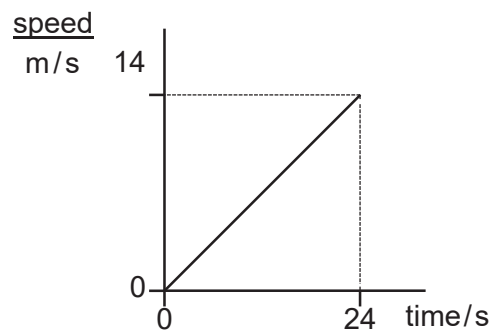
Question 3

Which distance/time graph represents the motion of an object moving at constant speed?



Question 4

The graph shows how the speed of a car changes with time.



Which calculation gives the distance travelled by the car in 24 seconds?

- A** $\left(\frac{14}{24}\right)\text{m}$
- B** $\left(\frac{24}{14}\right)\text{m}$
- C** $\left(\frac{24 \times 14}{2}\right)\text{m}$
- D** $(24 \times 14)\text{m}$

Question 5

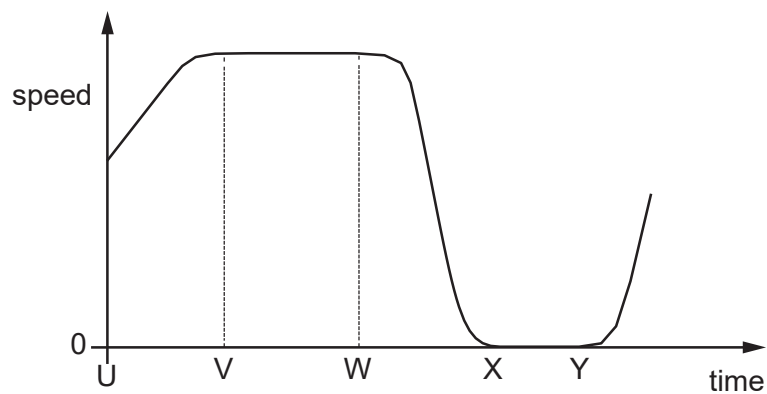
A car takes 15 minutes to travel along a road that is 20 km long.

What is the average speed of the car?

- A 0.75 km/h B 5.0 km/h C 80 km/h D 300 km/h

Question 6

The graph shows how the speed of a car changes with time.

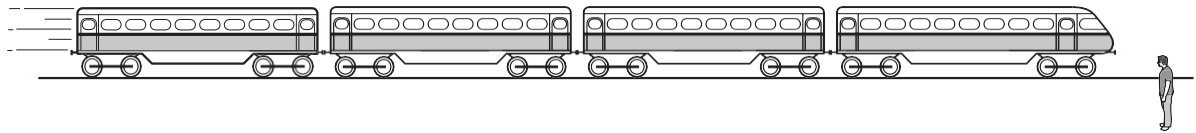


Between which two times is the car stationary?

- A U and V B V and W C W and X D X and Y

Question 7

A man stands by a railway track.



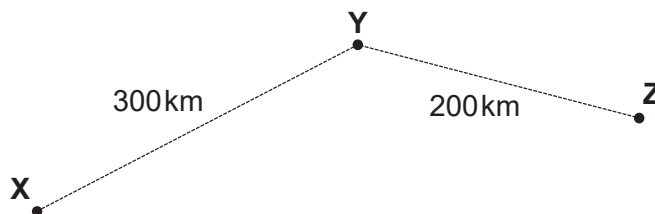
A train travelling at 40 m/s takes 2.0 s to pass the man.

What is the length of the train?

- A 20 m B 38 m C 40 m D 80 m

Question 8

An aeroplane flies from town **X** to town **Z**, stopping for 1 hour at town **Y** to pick up more passengers. The distances between the towns are shown in the diagram.



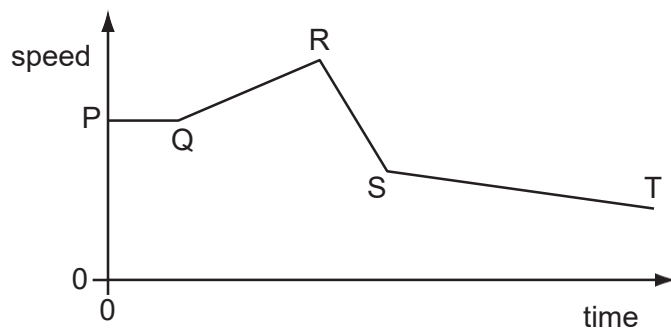
The total time taken between leaving **X** and arriving at **Z** is 3 hours.

What is the average speed of the aeroplane **in the air**?

- A $\frac{500}{4}$ km / h B $\frac{500}{3}$ km / h C $\frac{500}{2}$ km / h D $\frac{500}{1}$ km / h

Question 9

The diagram shows the speed/time graph for a train as it travels along a track.



For which part of the graph is the train's speed changing at the greatest rate?

A PQ

B QR

C RS

D ST

Question 10

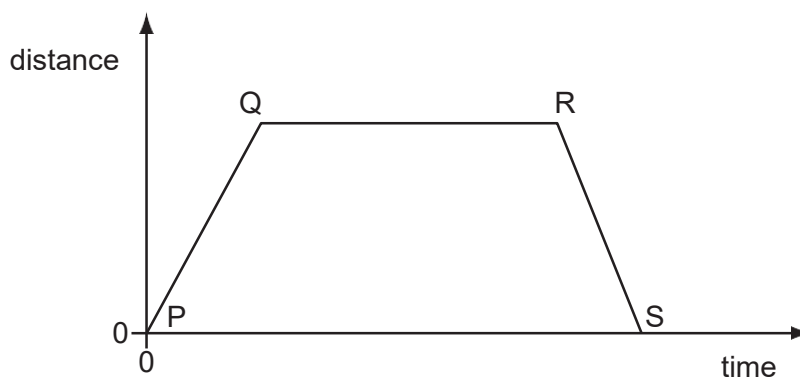
A small steel ball is dropped from a low balcony.

Ignoring air resistance, which statement describes its motion?

- A. It falls with constant acceleration.
- B. It falls with constant speed.
- C. It falls with decreasing speed.
- D. It falls with increasing acceleration.

Question 11

The graph shows how the distance travelled by a vehicle changes with time.



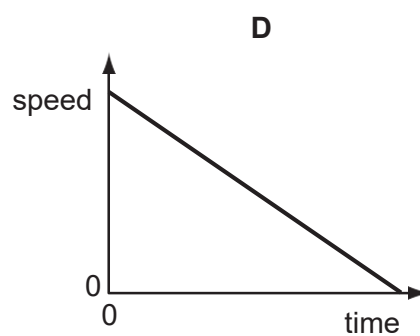
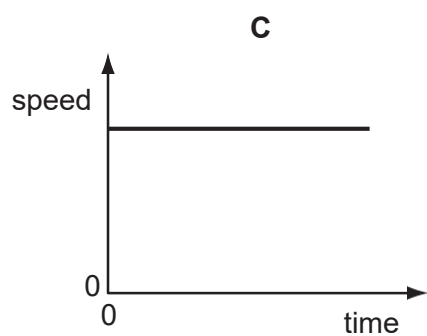
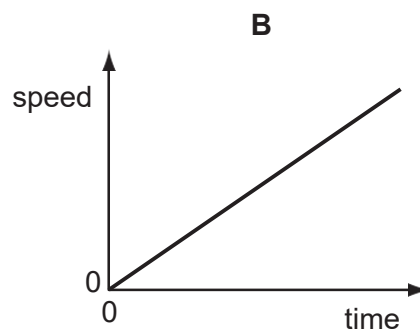
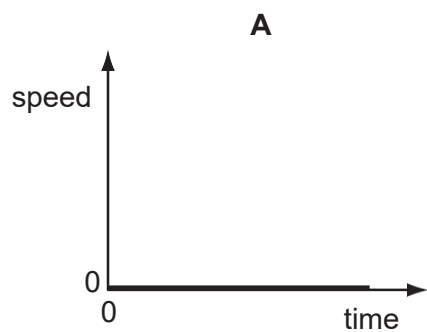
Which row describes the speed of the vehicle in each section of the graph?

	P to Q	Q to R	R to S
A	constant	zero	constant
B	constant	zero	decreasing
C	increasing	constant	decreasing
D	increasing	zero	decreasing

Question 12

A car is moving downhill along a road at a constant speed.

Which graph is the speed/time graph for the car?



Question 13

In a race, a car travels 60 times around a 3.6 km track. This takes 2.4 hours.

What is the average speed of the car?

- A 1.5 km/h B 90 km/h C 144 km/h D 216 km/h

Question 14

A tennis player hits a ball hard and 0.40s later hears the echo from a wall.



The speed of sound in air is 330 m/s.

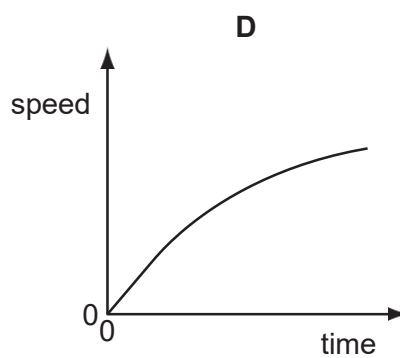
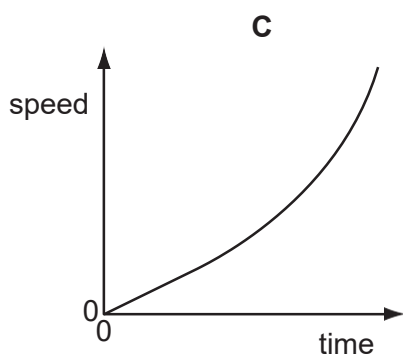
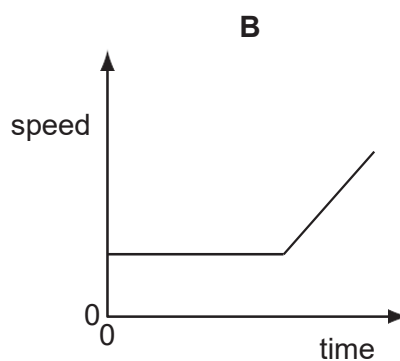
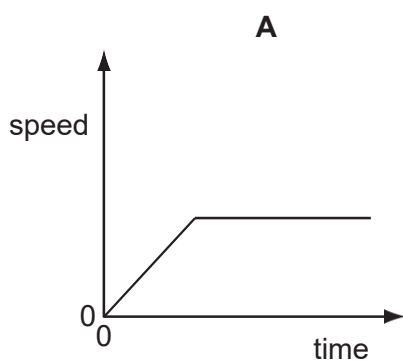
How far away is the player from the wall?

- A 66 m B 132 m C 264 m D 825 m

Question 15

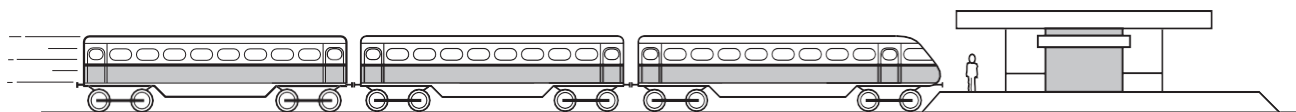
An object moves initially with constant speed and then with constant acceleration.

Which graph shows this motion?



Question 16

A child is standing on the platform of a station.



A train travelling at 30 m/s takes 3.0 s to pass the child.

What is the length of the train?

- A 10 m B 27 m C 30 m D 90 m

Mass & Weight

Question Paper 1

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Mass & Weight
Booklet	Question Paper 1

Time allowed: 18 minutes

Score: /14

Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

A cup contains hot liquid.

Some of the liquid evaporates.

What happens to the mass and what happens to the weight of the liquid in the cup?

	mass	weight
A	decreases	decreases
B	decreases	stays the same
C	stays the same	decreases
D	stays the same	stays the same

Question 2

An astronaut in an orbiting spacecraft experiences a force due to gravity. This force is less than when she is on the Earth's surface.

Compared with being on the Earth's surface, how do her mass and her weight change when she goes into orbit?

	mass in orbit	weight in orbit
A	decreases	decreases
B	decreases	unchanged
C	unchanged	decreases
D	unchanged	unchanged

Question 3

Diagram 1 shows a piece of foam rubber that contains many pockets of air. Diagram 2 shows the same piece of foam rubber after it has been compressed so that its volume decreases.

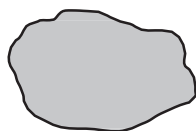


diagram 1
(before compression)

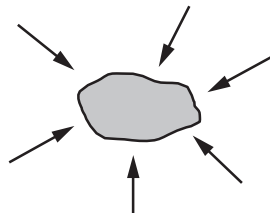


diagram 2
(after compression)

What happens to the mass and to the weight of the foam rubber when it is compressed?

	mass	weight
A	increases	increases
B	increases	no change
C	no change	increases
D	no change	no change

Question 4

Weight is an example of which quantity?

- A. acceleration
- B. force
- C. mass
- D. pressure

Question 5

The mass of an object is measured on Earth. The mass is 5.0 kg.

The object is taken to the Moon. The mass of the object is measured on the Moon.

What is the mass of the object on the Moon?

- A. 0 kg
- B. more than 0 kg, but less than 5.0 kg
- C. 5.0 kg
- D. more than 5.0 kg

Question 6

Which statement about mass or weight is correct?

- A. Mass is a force.
- B. Mass is measured in newtons.
- C. Weight is a force.
- D. Weight is measured in kilograms.

Question 7

What is the weight of an object?

- A. the force of gravity on the object
- B. the gravitational potential energy of the object
- C. the internal energy of the object
- D. the mass of the object

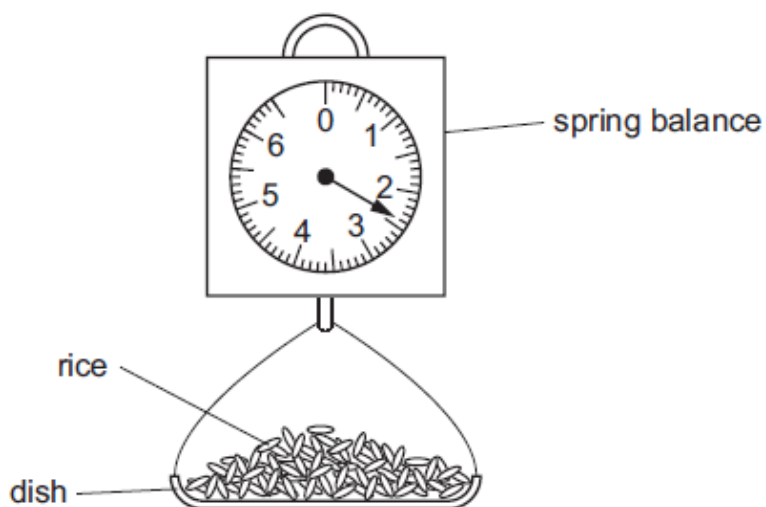
Question 8

Which instrument is used to compare the masses of objects?

- A. a balance
- B. a barometer
- C. a manometer
- D. a measuring cylinder

Question 9

A customer goes to a market and buys some rice. The stallholder pours rice into a dish that hangs from a spring balance. He records the reading on the spring balance.



The customer then buys some pasta and the stallholder notices that the reading on the spring balance, with just pasta in the dish, is the same as it was with just rice in the dish.

The rice and the pasta must have the same

- A. density.
- B. temperature.
- C. volume.
- D. weight.

Question 10

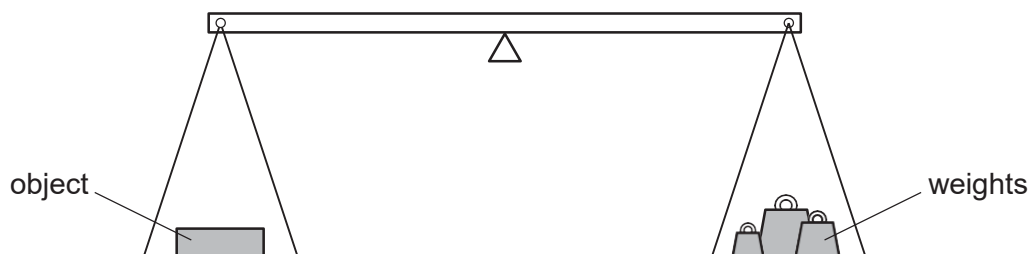
The mass of an astronaut is 70kg on the Moon.

What is the mass of the astronaut on the Earth?

- A 7 kg B 70 kg C 80kg D 700kg

Question 11

The weight of an object is found using the balance shown in the diagram. The object is put in the left-hand pan and various weights are put in the right-hand pan.



These are the results.

weights in the right-hand pan	effect
0.1 N, 0.1 N, 0.05 N, 0.02 N	balance tips down slightly on the left-hand side
0.2 N, 0.1 N, 0.01 N	balance tips down slightly on the right-hand side

What is the best estimate of the weight of the object?

- A 0.27 N B 0.29 N C 0.31 N D 0.58 N

Question 12

Which is the unit for force and which is the unit for weight?

	force	weight
A	kg	kg
B	kg	N
C	N	kg
D	N	N

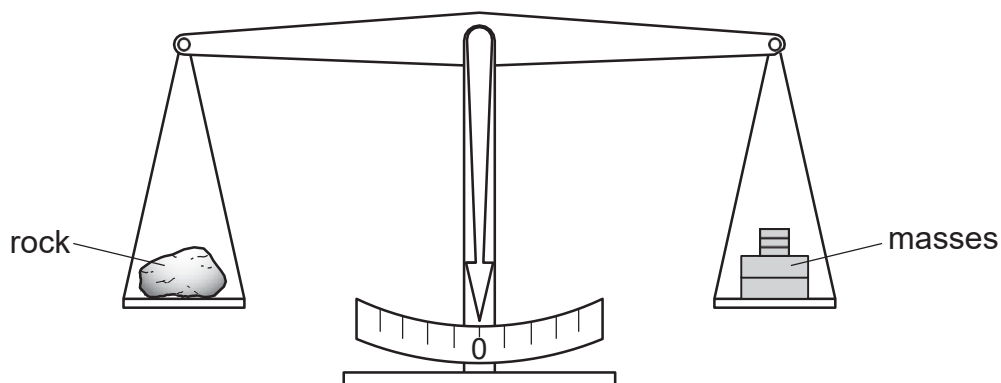
Question 13

Which quantity is measured in newtons?

- A. density
- B. energy
- C. pressure
- D. Weight.

Question 14

A geologist places a small rock on the left-hand pan of a balance. The two pans are level as shown when masses with a total weight of 23 N are placed on the right-hand pan. Take the weight of 1.0 kg to be 10 N.



What is the mass of the small rock?

- A 0.023 kg B 2.3 kg C 23 kg D 230 kg

Density

Question Paper 1

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Density
Booklet	Question Paper 1

Time allowed: 20 minutes

Score: /16

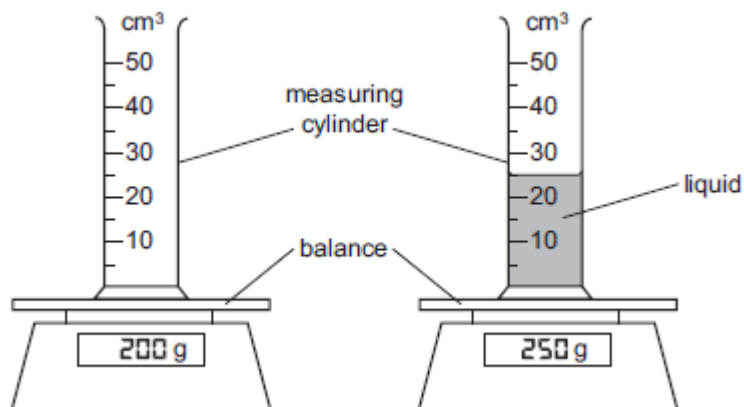
Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

The diagram shows an experiment to find the density of a liquid.

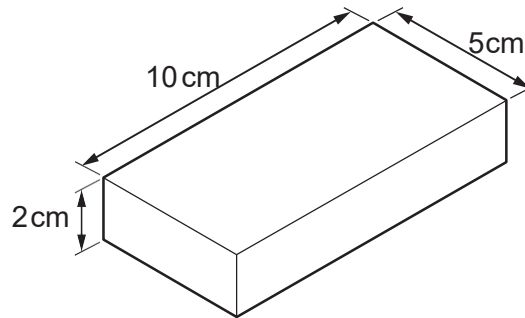


What is the density of the liquid?

- A 0.5g/cm³ B 2.0g/cm³ C 8.0g/cm³ D 10.0g/cm³

Question 2

A metal block has the dimensions shown. Its mass is 1000g.

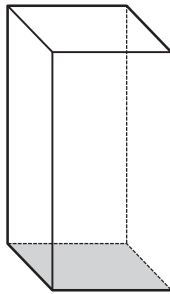


What is the density of the metal?

- A** $\left(\frac{5 \times 10}{1000 \times 2} \right) \text{g/cm}^3$
- B** $\left(\frac{2 \times 5 \times 10}{1000} \right) \text{g/cm}^3$
- C** $\left(\frac{1000 \times 2}{5 \times 10} \right) \text{g/cm}^3$
- D** $\left(\frac{1000}{2 \times 5 \times 10} \right) \text{g/cm}^3$

Question 3

A student wishes to determine the density of the solid block shown.



Which quantities must be known?

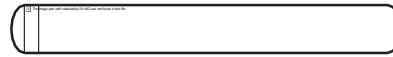
- A the area of the shaded face and the volume of the block
- B the area of the shaded face and the weight of the block
- C the mass of the block and the height of the block
- D the mass of the block and the volume of the block

Question 4

Two cylinders are made of the same metal. Both cylinders have the same cross-sectional area but one is longer than the other.



cylinder 1



cylinder 2

Which quantity is the same for both cylinders?

- A. density
- B. mass
- C. resistance
- D. volume

Question 5

The mass of a piece of metal is 1200g.

A measuring cylinder contains 150 cm³ of water.

The piece of metal is put into the measuring cylinder. The water level rises to 250 cm³ and covers the metal.

What is the density of the metal?

- A 3.0g/cm³ B 4.8g/cm³ C 8.0g/cm³ D 12.0g/cm³

Question 6

A person measures the length, width, height and mass of a metal block with rectangular sides.

Which of these measurements must be used in order to calculate the density of the metal?

- A. mass only
- B. height and mass only
- C. length, width and height only
- D. length, width, height and mass

Question 7

A liquid has a volume of 100 cm^3 and a mass of 85 g .

The density of water is 1.0 g/cm^3 .

How does the density of the liquid compare with the density of water?

- A Its density is higher than that of water.
- B Its density is lower than that of water.
- C Its density is the same as that of water.
- D It is impossible to say with only this data.

Question 8

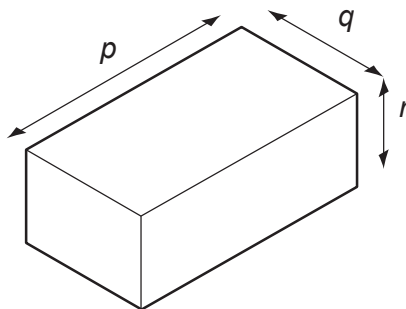
The table gives the volumes and masses of four objects.

Which object has the greatest density?

	mass/g	volume /cm ³
A	5.4	2.0
B	13	3.0
C	15	6.0
D	18	5.0

Question 9

The diagram shows the dimensions of a rectangular block of metal of mass m .



Which expression is used to calculate the density of the metal?

- A. $m \times p \times q$
- B. $m \times p \times q \times r$
- C. $\frac{m}{(p \times q)}$
- D. $\frac{m}{(p \times q \times r)}$

Question 10

Diagram 1 shows an empty measuring cylinder on a balance.

Diagram 2 shows the same measuring cylinder on the balance, but it now contains a liquid.

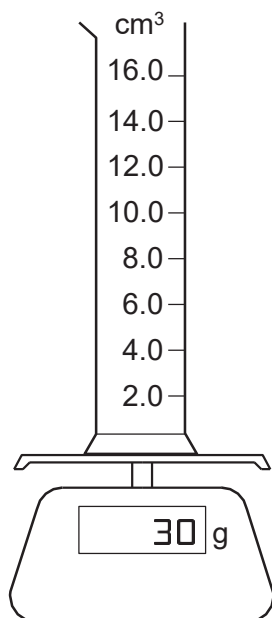


diagram 1

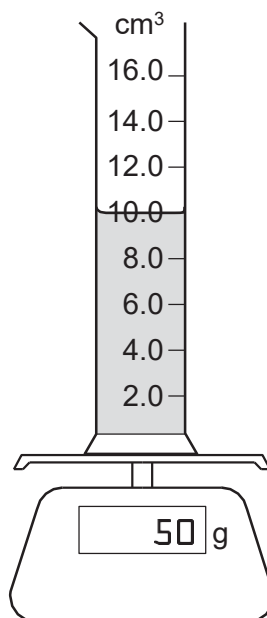


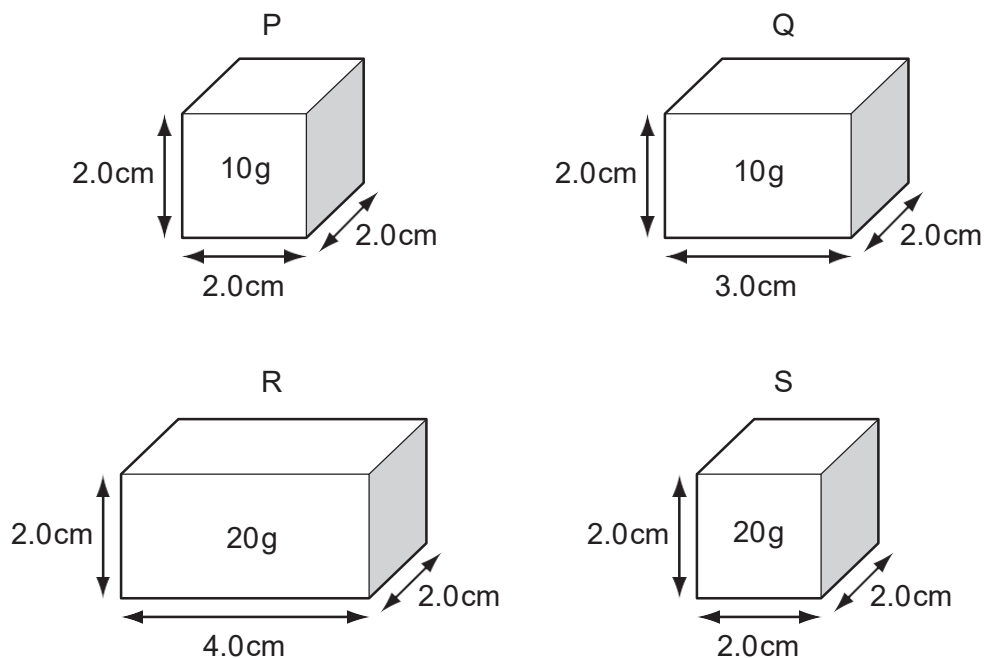
diagram 2

What is the density of the liquid?

- A 0.2 g/cm³ B 0.5 g/cm³ C 2.0 g/cm³ D 5.0 g/cm³

Question 11

Four rectangular blocks, P, Q, R and S are shown. Each block is labelled with its size and its mass.



Which two blocks have the same density?

- A P and Q B P and R C Q and R D R and S

Question 12

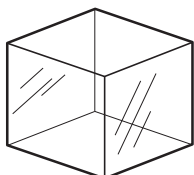
A student is given four different objects and a metre rule.

Each object has a known mass. She is asked to determine the densities of the materials from which the four objects are made.

The objects are a copper cylinder, a glass cube, a steel spanner and a stone tile.



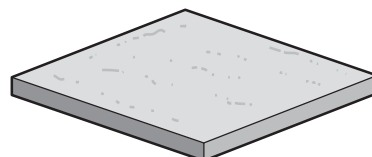
copper



glass



steel



stone

Using only the metre rule, she is able to find the densities of only three of the four materials.

Which three materials are these?

- A copper, glass and steel
- B copper, glass and stone
- C copper, steel and stone
- D glass, steel and stone

Question 13

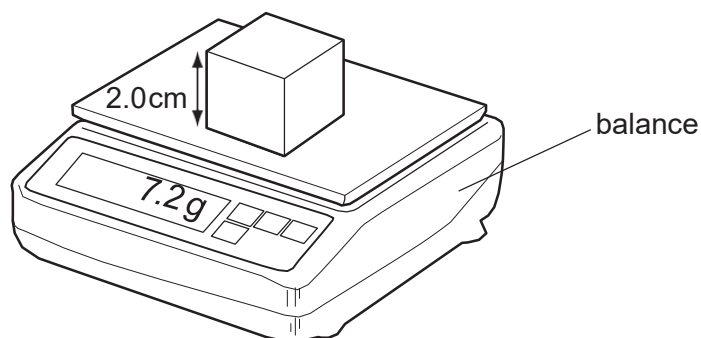
A stone has a volume of 0.50 cm^3 and a mass of 2.0 g .

What is the density of the stone?

- A 0.25 g/cm^3
- B 1.5 g / cm^3
- C 2.5 g / cm^3
- D 4.0 g/cm^3

Question 14

A cube of side 2.0cm is placed on a balance.



What is the density of the cube?

- A 0.90g/cm^3 B 1.2g/cm^3 C 1.8g/cm^3 D 3.6g/cm^3

Question 15

A student is told to measure the density of a liquid and also of a large cube of metal.

Which pieces of equipment are sufficient to be able to take the measurements needed?

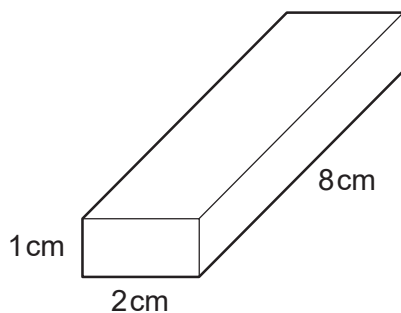
- A. balance, measuring cylinder and ruler
- B. balance and thermometer
- C. measuring cylinder and ruler
- D. measuring cylinder, ruler and thermometer

Question 16

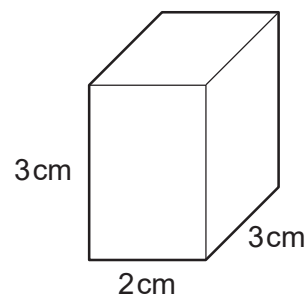
The diagrams show four blocks with the same mass.

Which block is made from the least dense material?

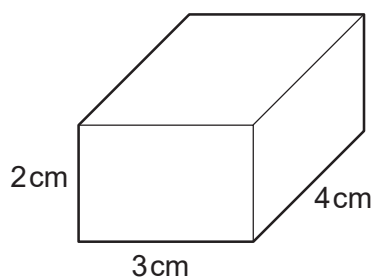
A



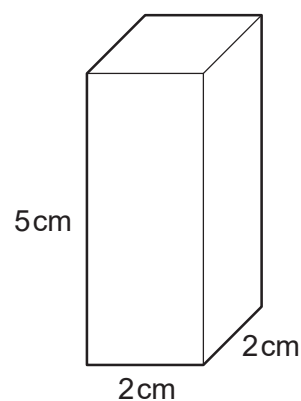
B



C



D



F = ma / Resultant Forces

Question Paper 1

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	F = ma / Resultant Forces
Booklet	Question Paper 1

Time allowed: 23 minutes

Score: /18

Percentage: /100

Grade Boundaries:

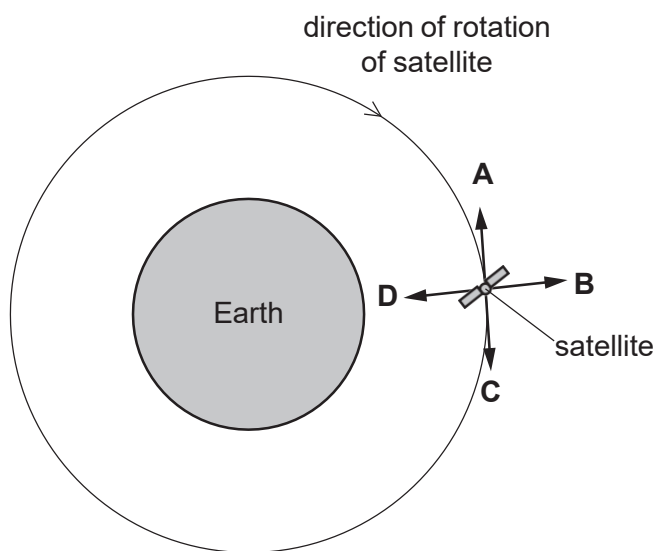
9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

A satellite orbits the Earth above the atmosphere at a constant speed.

The diagram shows the satellite at one point in its circular orbit around the Earth.

Which labelled arrow shows the direction of the resultant force on the satellite at the position shown?

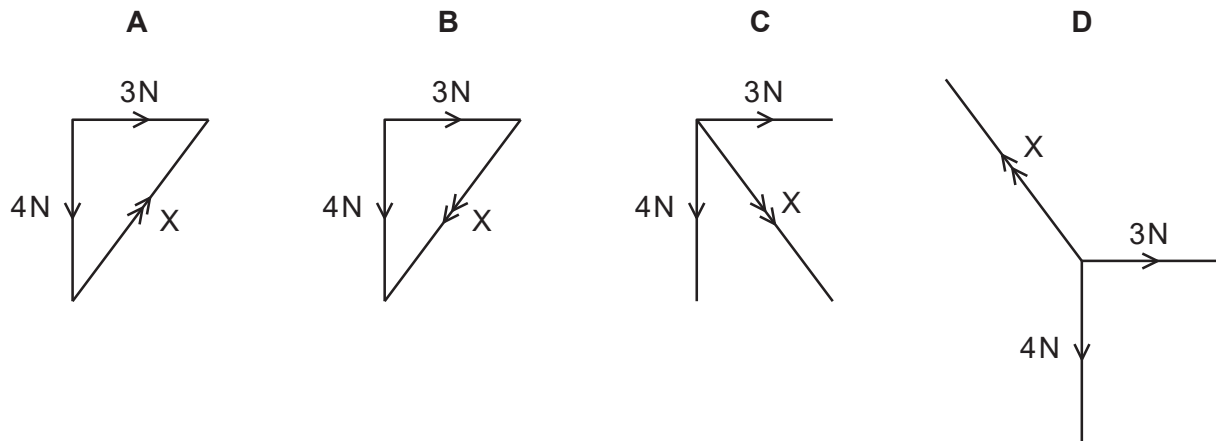


Question 2

An object is acted upon by a 3 N force and by a 4 N force.

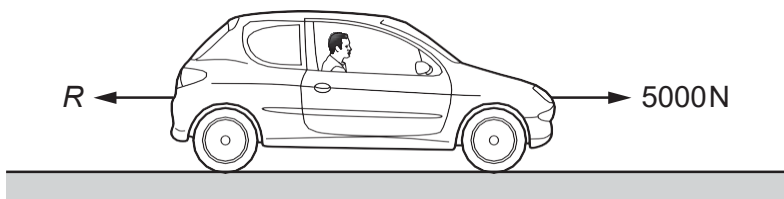
Each diagram shows the two forces.

Which diagram also shows the resultant X of these two forces?



Question 3

The engine of a car produces a driving force of 5000 N on the car. Resistive forces R also act on the car, as shown.



The car has a mass of 800 kg and an acceleration of 1.0 m/s^2 .

What is the value of R ?

- A 800 N B 4200 N C 5800 N D 8000 N

Question 4

An object moves in a circle at constant speed.

Which statement about the force needed on the object is correct?

- A. A force away from the centre of the circle keeps the object moving in the circle.
- B. A force in the direction of motion of the object keeps it moving in the circle.
- C. A force towards the centre of the circle keeps the object moving in the circle.
- D. No force is needed to keep the object moving at constant speed in the circle.

Question 5

A parachutist is falling at terminal velocity, without her parachute open.

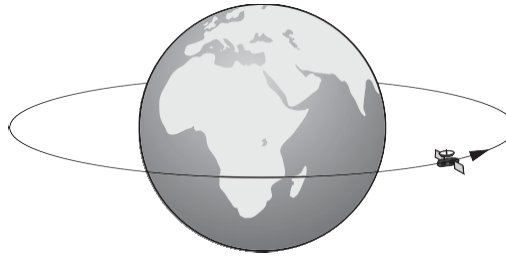
She now opens her parachute.

What is the direction of her motion, and what is the direction of her acceleration, immediately after she opens her parachute?

	direction of motion of the parachutist	direction of acceleration of the parachutist
A	downwards	downwards
B	downwards	upwards
C	upwards	downwards
D	upwards	upwards

Question 6

The diagram shows a satellite that is moving at a uniform rate in a circular orbit around the Earth.



Which statement describes the motion of this satellite?

- A It is accelerating because its speed is changing.
- B It is accelerating because its velocity is changing.
- C It is not accelerating but its speed is changing.
- D It is not accelerating but its velocity is changing.

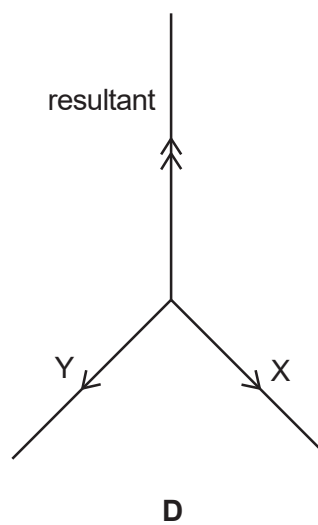
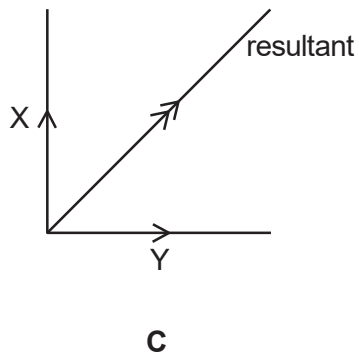
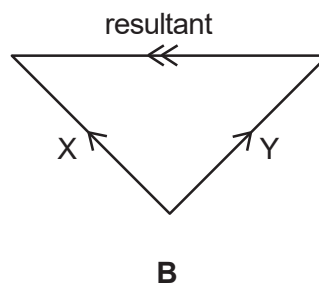
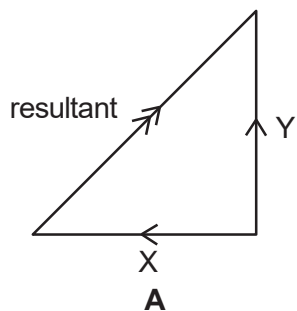
Question 7

Which statement about an object moving in a straight line through air is correct?

- A. When it accelerates, the resultant force acting on it is zero.
- B. When it moves at a steady speed, the air resistance acting on it is zero.
- C. When it moves at a steady speed, the resultant force acting on it is zero.
- D. When it moves, there is a resultant force acting on it.

Question 8

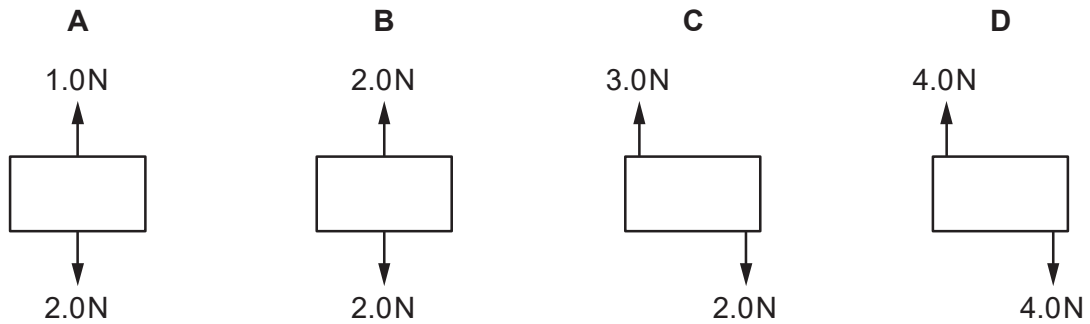
Which diagram shows two forces X and Y with their resultant force?



Question 9

Four objects are each acted on by only two forces, as shown.

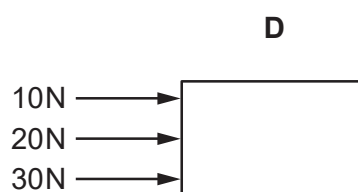
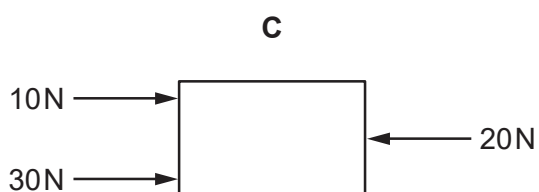
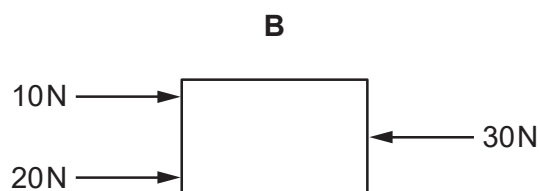
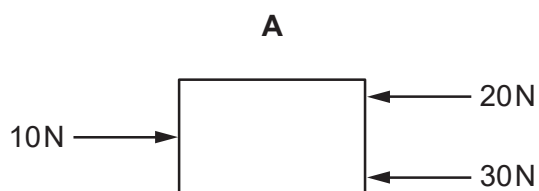
Which object is in equilibrium?



Question 10

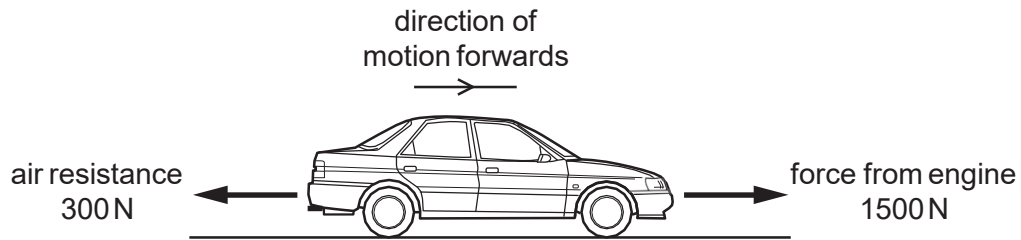
The diagrams show four identical objects. Each object is acted on by only the three forces shown.

Which object accelerates to the right, with the **smallest** acceleration?



Question 11

A car travels along a horizontal road at a constant speed. Three horizontal forces act on the car. The diagram shows two of these three forces.



What is the size and the direction of the third horizontal force acting on the car?

- A 1200N backwards
- B 1200 N forwards
- C 1800N backwards
- D 1800 N forwards

Question 12

Which list contains only properties of an object that can be changed by a force?

- A direction of motion, mass, shape
- B direction of motion, mass, speed
- C direction of motion, shape, speed
- D mass, shape, speed

Question 13

The diagram shows the only three forces acting on an object.



What is the resultant force on the object?

- A. 0 N
- B. 5.0 N towards the left
- C. 5.0 N towards the right
- D. 10.0 N towards the right

Question 14

In which situation is no resultant force needed?

- A. a car changing direction at a steady speed
- B. a car moving in a straight line at a steady speed
- C. a car slowing down
- D. a car speeding up.

Question 15

Which properties of a body can be changed by applying a force to the body?

- A. mass, motion and shape
- B. mass and motion, but not shape
- C. mass and shape, but not motion
- D. motion and shape, but not mass

Question 16

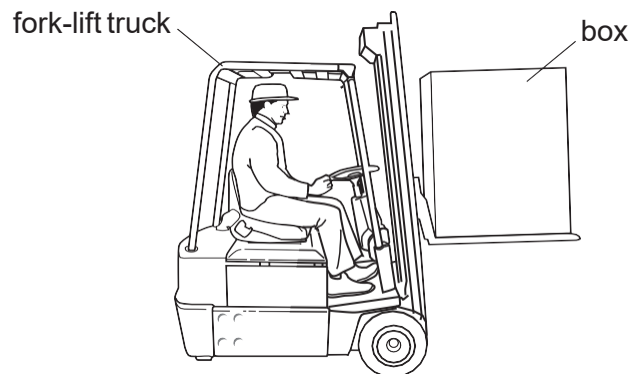
A force acting on an object causes some properties of the object to change.

Which list contains **only** properties that can be changed by the action of the force?

- A mass, motion and shape
- B mass, motion and size
- C mass, shape and size
- D motion, shape and size

Question 17

A box is being moved by a fork-lift truck. The total weight of the box is 3000 N.



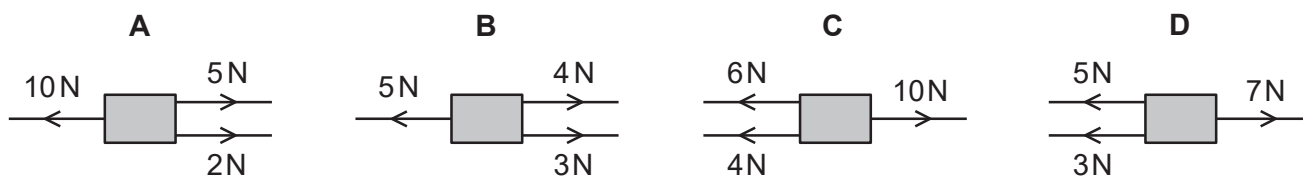
The force exerted by the fork-lift truck on the box is 3500 N upwards.

What is the resultant force on the box?

- A. 500 N downwards
- B. 500 N upwards
- C. 6500 N downwards
- D. 6500 N upwards

Question 18

Which combination of forces produces a resultant force acting towards the right?



Moments / Centre of Mass

Question Paper 1

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Moments / Centre of Mass
Booklet	Question Paper 1

Time allowed: 19 minutes

Score: /15

Percentage: /100

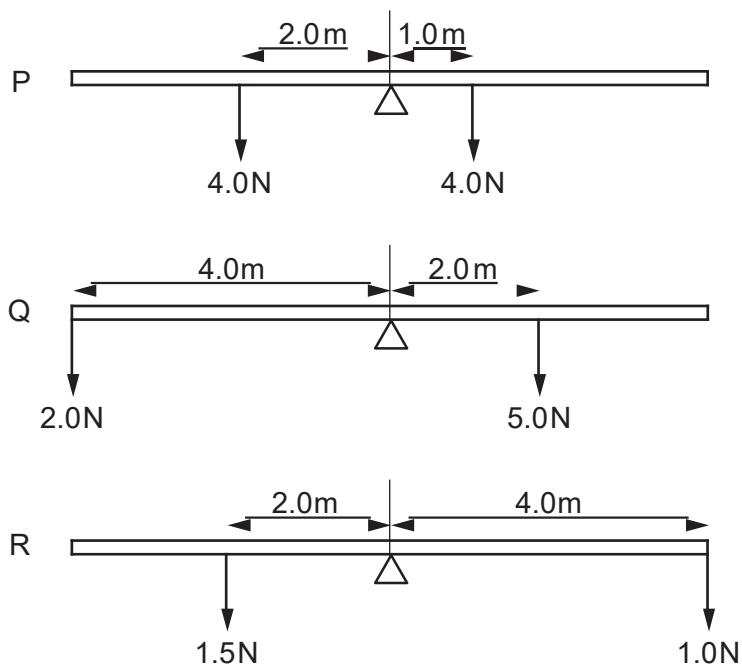
Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

The diagrams show three uniform beams P, Q and R, each pivoted at its centre.

The two forces acting on each beam are also shown.

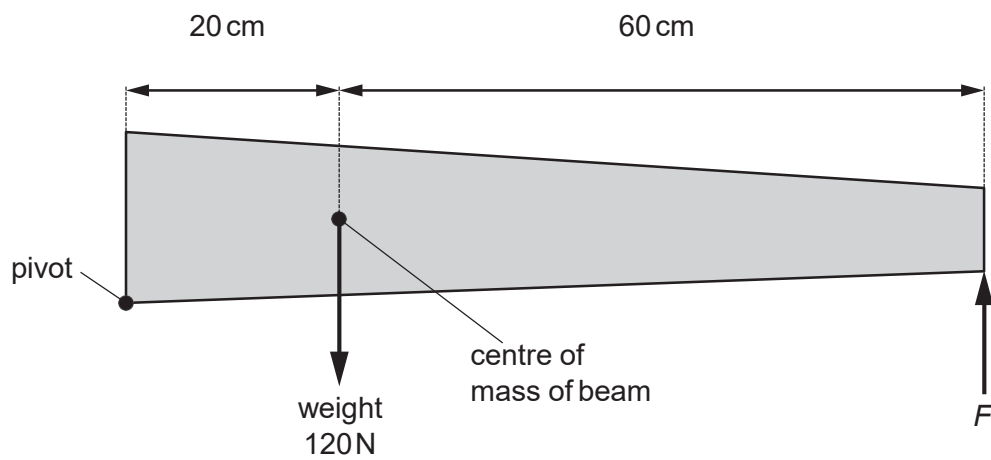


Which beams rotate clockwise?

- A. P and Q only
- B. P and R only
- C. Q and R only
- D. P, Q and R

Question 2

The diagram shows a non-uniform beam of weight 120 N, pivoted at one end. The beam is kept in equilibrium by force F .



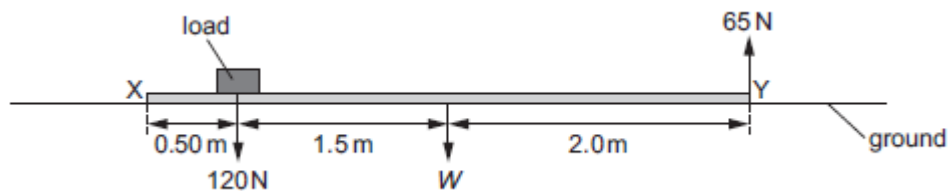
What is the value of force F ?

- A 30 N B 40 N C 360 N D 480 N

Question 3

A long plank XY lies on the ground. A load of 120 N is placed on it, at a distance of 0.50 m from end X, as shown.

End Y is lifted off the ground. The upward force needed to do this is 65 N.



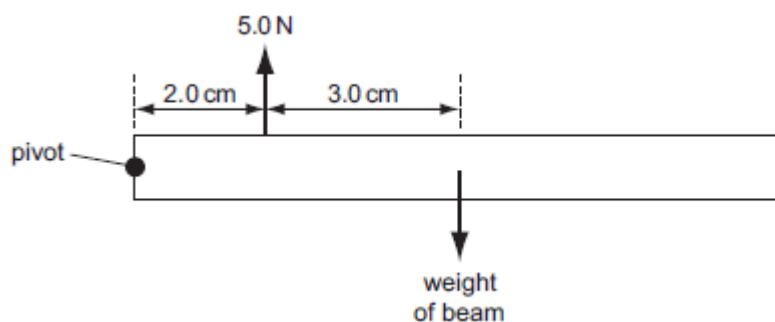
In the diagram, W is the weight of the plank, acting at its mid-point.

What is the value of W ?

- A 35 N B 47 N C 100 N D 133 N

Question 4

A beam pivoted at one end has a force of 5.0 N acting vertically upwards on it as shown. The beam is in equilibrium.



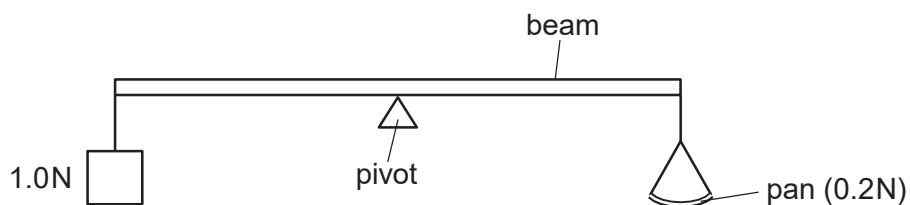
What is the weight of the beam?

- A 2.0 N B 3.0 N C 3.3 N D 5.0 N

Question 5

The diagram shows a uniform beam being used as a balance. The beam is pivoted at its centre.

A 1.0 N weight is attached to one end of the beam. An empty pan weighing 0.2 N is attached to the other end of the beam.

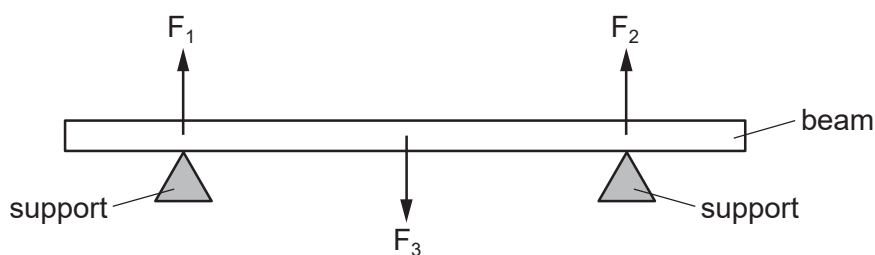


How many 0.1 N weights must be placed on the pan in order to balance the beam?

- A 5 B 8 C 10 D 12

Question 6

A heavy beam rests on two supports. The diagram shows the only three forces F_1 , F_2 and F_3 acting on the beam.



The beam is in equilibrium.

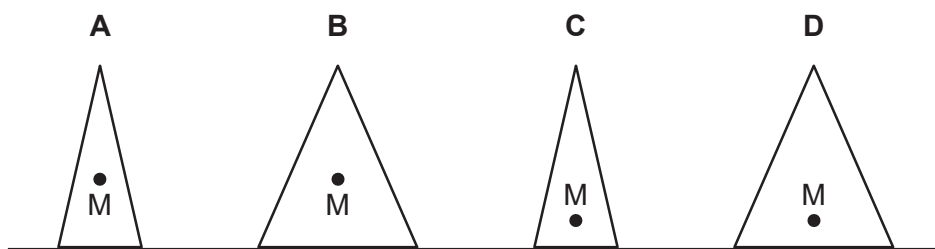
Which statement is correct?

- A. All the forces are equal in size.
- B. The resultant force on the beam is in the opposite direction to the resultant turning effect.
- C. The resultant force on the beam is zero and the resultant turning effect on the beam is zero.
- D. The total upward force is twice the total downward force.

Question 7

The diagrams show four solid cones. The centre of mass of each cone is marked by a point labelled M.

Which cone is the most stable?



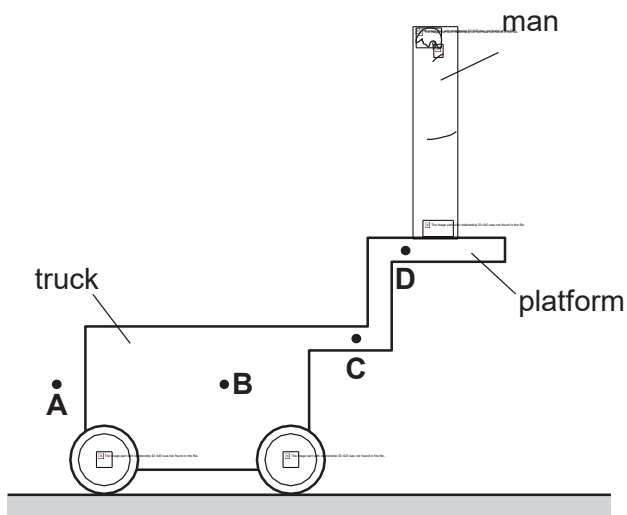
Question 8

A heavy truck on wheels has a platform attached to it.

A man stands on the platform.

The truck does not fall over.

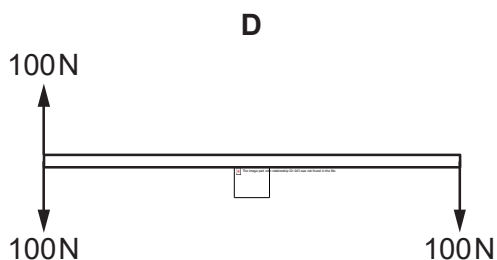
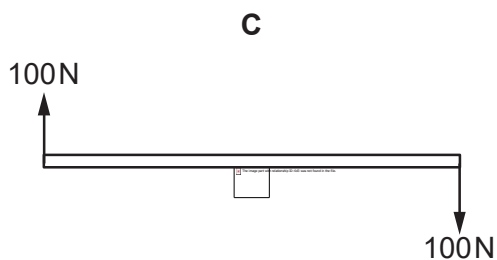
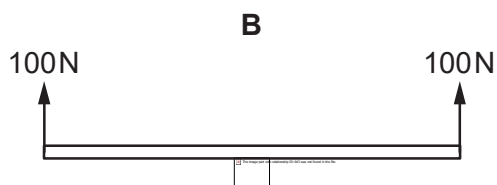
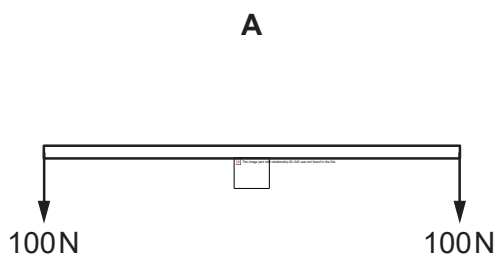
Which position **A**, **B**, **C** or **D** could be the centre of mass of the whole system (truck, platform and man)?



Question 9

A uniform rod rests on a pivot at its centre. The rod is not attached to the pivot. Forces are then applied to the rod in four different ways, as shown. The weight of the rod can be ignored.

Which diagram shows the rod in equilibrium?

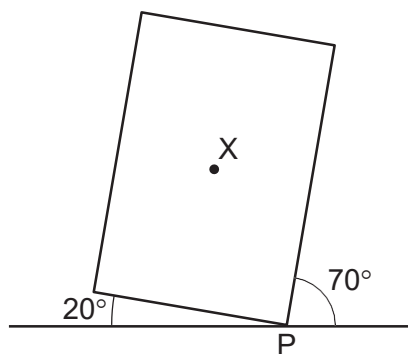


Question 10

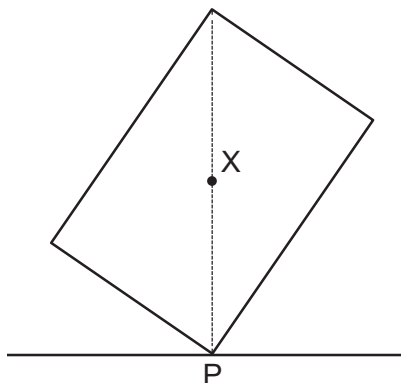
A plane lamina with centre of mass X touches the ground at point P .

Which diagram shows the lamina in equilibrium?

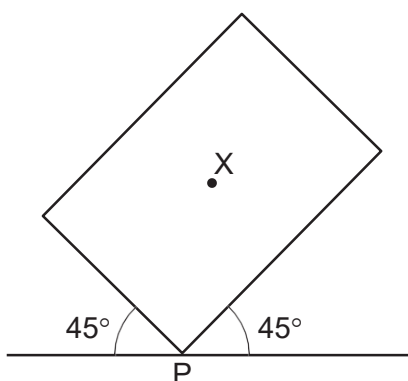
A



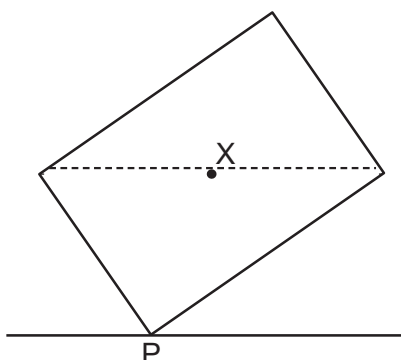
B



C



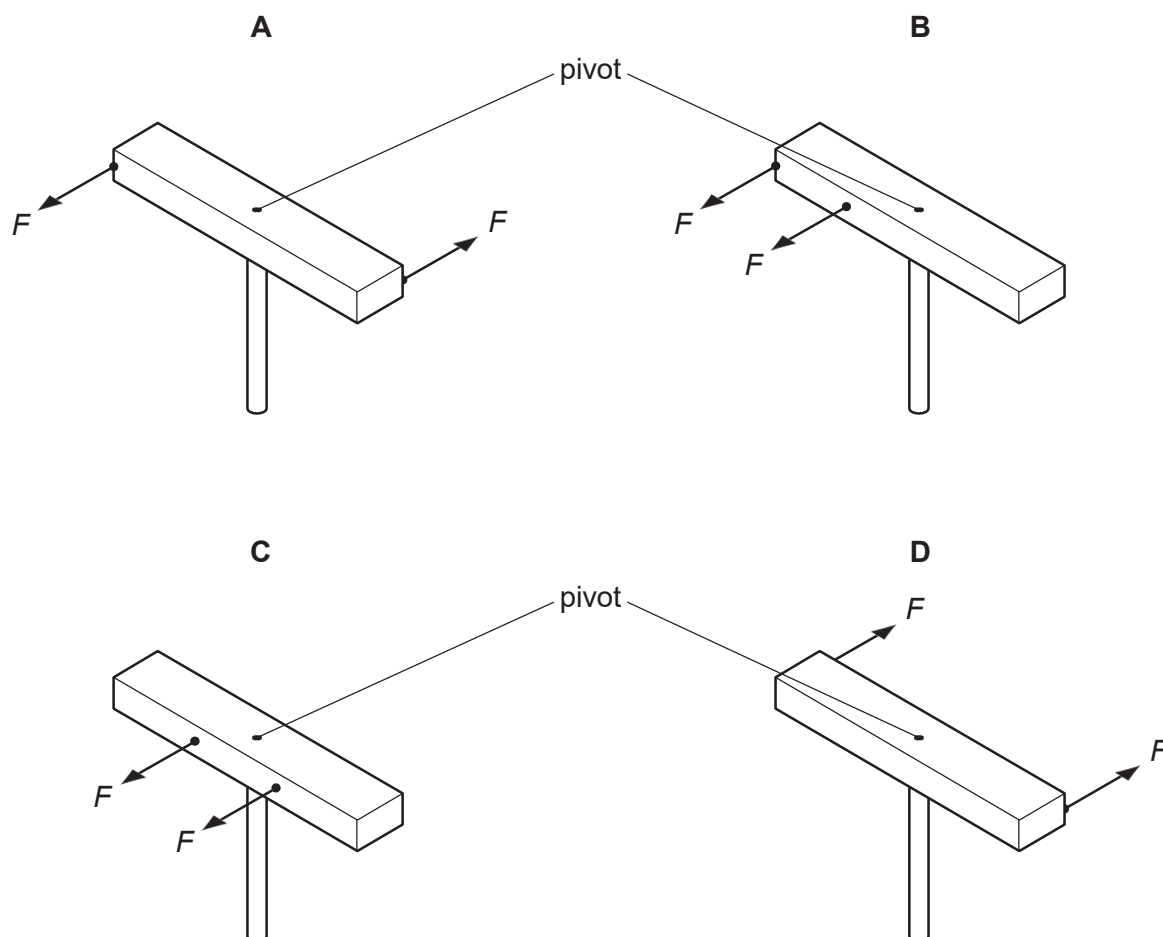
D



Question 11

A wooden bar is pivoted at its centre so that it can rotate freely. Two equal forces F are applied to the bar.

In which diagram is the turning effect greatest?

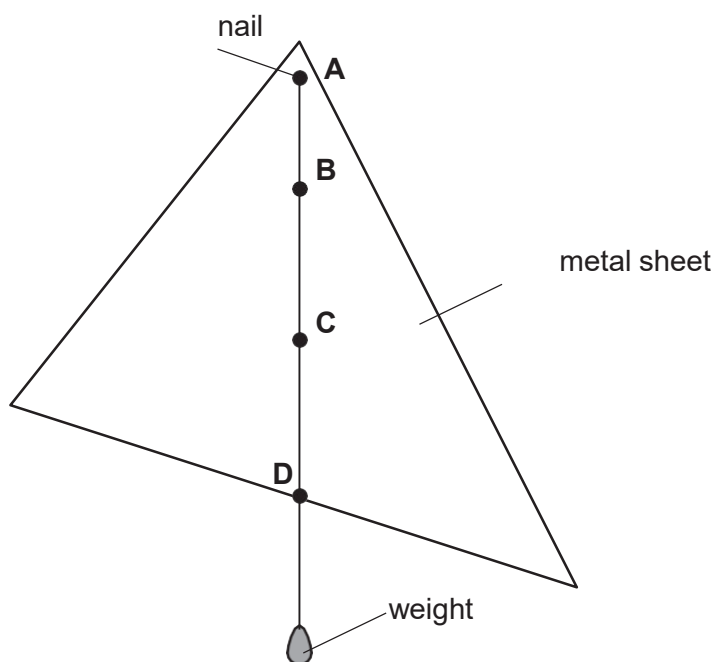


Question 12

The diagram shows a uniform, flat metal sheet hanging freely from a nail at point **A**. A weight also hangs freely on a string tied to **A**.

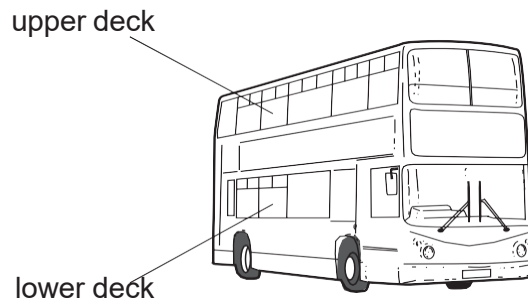
One of the labelled points is at the centre of mass of the metal sheet.

Which point is at the centre of mass?



Question 13

Passengers are not allowed to stand on the upper deck of double-decker buses.

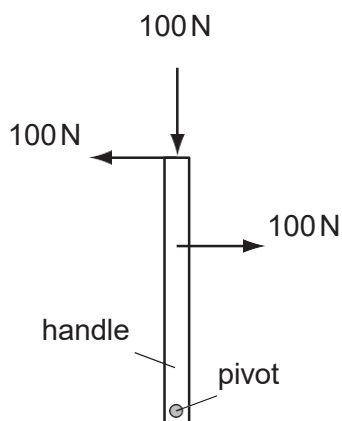


Why is this?

- A. They would cause the bus to become less stable.
- B. They would cause the bus to slow down.
- C. They would increase the kinetic energy of the bus.
- D. They would lower the centre of mass of the bus.

Question 14

The diagram shows a handle with three forces, each 100 N, applied to it. The handle is free to move.



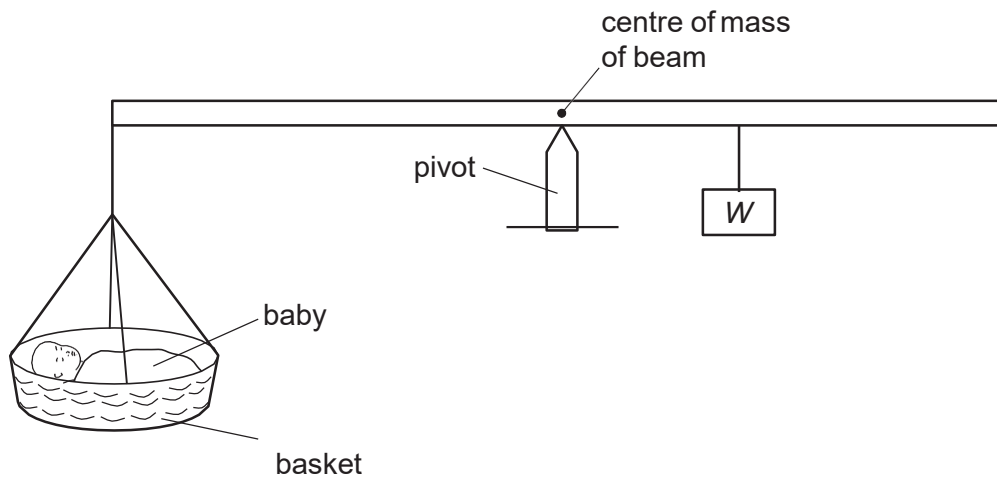
What is the effect of the forces on the handle?

- A. The handle will move downwards.
- B. The handle will not move.
- C. The handle will turn anticlockwise (to the left).
- D. The handle will turn clockwise (to the right).

Question 15

The diagram shows a balance being used to find the weight of a baby. The weight of the basket can be ignored.

At equilibrium, the pivot is nearer to the weight W than to the baby.



What is the weight of the baby?

- A less than W
- B more than W
- C W
- D impossible to tell

1. Nov/2023/Paper_0625/21/No.7

A sphere X collides head on with a second identical sphere Y which is stationary.

The mass of each sphere is 0.15 kg.

Sphere X is travelling at a velocity of 2.0 m/s before the collision and produces an impulse of 0.21 N s on sphere Y.

What is the velocity of sphere X after collision?

A 0.60 m/s in the opposite direction to Y

☒ B 0.60 m/s in the same direction as Y

C 1.4 m/s in the opposite direction to Y

D 1.4 m/s in the same direction as Y

Impulse = $f \times t = mv - mu$ for Y $mu = 0$
 $mv = 0.21 \text{ N s}$

Diagram showing spheres X and Y before and after collision. Before: X (0.15 kg, 2.0 m/s) and Y (0.15 kg, 0). After: X (0.15 kg, v) and Y (0.15 kg, 0.21).
 $(0.15 \times 2) + (0.15 \times 0) = (0.15 \times v) + 0.21$
 $3 + 0 = 0.15v + 0.21$
 $0.15v = 3 - 0.21$
 $v = \frac{2.79}{0.15} = 18.6 \text{ m s}^{-1}$

2. Nov/2023/Paper_0625/22/No.7

A resultant force F accelerates a car of mass m along a straight horizontal road from rest to a speed v in time t , giving it momentum p .

Which pair of relationships for this situation is correct?

A $pt = mv$ and $F = pt$

B $p = mv$ and $F = pt$

☒ C $p = mv$ and $Ft = p$

D $p = mvt$ and $Ft = v$

$F = m \times a$
 $a = \frac{v - u}{t}$ but $u = 0$
 $a = \frac{v}{t}$
 $p = m \times v = mv$
 $F = \frac{m \times v}{t}$
 $F \times t = mv$
 $Ft = p$

3. Nov/2023/Paper_0625/23/No.7

A car of mass 1200 kg is travelling along a straight horizontal road.

Which impulse is needed to accelerate the car from 5.0 m/s to 10 m/s?

☒ A 6000 N s

B 12000 N s

C 15000 N s

D 18000 N s

Impulse = $mv - mu$
 $= m(v - u)$
 $= 1200(10 - 5)$
 $= 1200 \times 5$
 $= 6000 \text{ N s}$

- (a) A balloon of mass 15 g is glued to a straw. The straw is threaded onto a horizontal string, as shown in Fig. 3.1. $\uparrow 0.015\text{ kg}$
The balloon is filled with air and then the air is released.

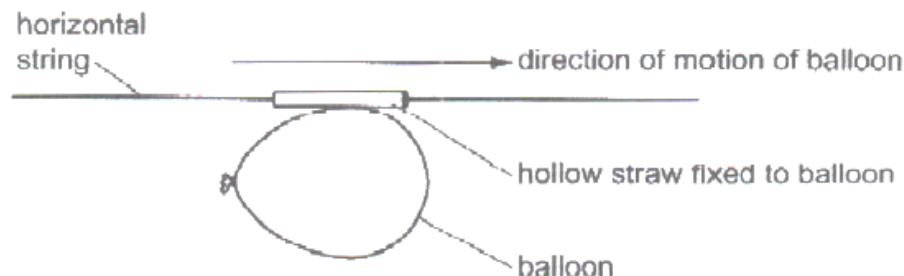


Fig. 3.1

As the air leaves the balloon, the balloon experiences a force.

The balloon accelerates from rest until it reaches a constant speed. It then travels 0.67 m in 0.18 s at this constant speed.

- (i) Explain in words what is meant by the term impulse.

Is force multiplied by time for which the force acts.

[1]

- (ii) Calculate the resultant impulse on the balloon while it is accelerating.

$$\text{Impulse} = F \times t$$

$$F \times t = mv - mu$$

$$= 0.015 \left(\frac{0.67}{0.18} - 0 \right)$$

$$= 0.0558\text{ N s}$$

$$\approx 0.056\text{ N s} \quad \text{impulse} = 0.056\text{ N s}$$

[3]

- (iii) Explain how momentum is conserved as the balloon accelerates.

- Released air is in opposite direction to balloon motion.

- Momentum of balloon and straw is equal in size (magnitude) to the momentum of air.

[2]

Fig. 3.1 shows a boy throwing a ball at an object in a fairground.



Fig. 3.1

The ball has a mass of 190g and travels horizontally with a constant speed of 6.9 m/s.

- (a) Calculate the momentum of the ball.

$$\begin{aligned}
 p &= m \times v \\
 &= 0.19 \times 6.9 \\
 &= 1.311 \text{ kg m/s} \\
 &\approx 1.3 \text{ kg m/s}
 \end{aligned}$$

momentum = 1.3 [2]

- (b) After hitting the object, the ball bounces back along the same straight path with a speed of 1.5 m/s. The object has a mass of 1.8 kg.

↑ velocity in opposite direction is -ve. (velocity is a vector).
Calculate the speed of the object after it is hit by the ball.

$$\begin{aligned}
 p \text{ before} &= p \text{ after collision} \\
 1.3 + 0 &= (0.19 \times -1.5) + (1.8 \times v)
 \end{aligned}$$

$$1.3 = -0.285 + 1.8v$$

$$v = \frac{1.3 + 0.285}{1.8}$$

$$= 0.88 \text{ m/s}$$

speed = 0.88 m/s [3]

(c) The kinetic energy of the ball is 4.5 J before the collision and 0.2 J after the collision.

Calculate the change in total kinetic energy of the ball and object during the collision.

$K.E \text{ before} - K.E \text{ after}$

$$K.E = \frac{1}{2}mv^2$$

$$4.5 - (0.2 + \frac{1}{2} \times 1.8 \times 0.38^2)$$

$$4.5 - (0.2 + 0.696)$$

$$4.5 - 0.896 = \underline{3.6} \text{ J}$$

change in total kinetic energy = 3.6 J [3]

[Total: 8]

6. June/2023/Paper_0625/21/No.9

An object of mass 1.2 kg is moving with a velocity of 2.0 m/s when it is acted on by a force of 4.0 N. The velocity of the object increases to 5.0 m/s in the same direction.

For which period of time does the force act on the object?

- (A) 0.90 s B 1.1 s C 1.5 s D 3.6 s

$$F = m \times a$$

$$a = \frac{F}{m}$$

$$= \frac{4}{1.2}$$

$$a = 3.3 \text{ m/s}^2$$

$$a = \frac{v - u}{t}$$

$$t = \frac{v - u}{a}$$

$$u = 2.0 \text{ m/s}$$

$$v = 5.0 \text{ m/s}$$

$$t = \frac{5.0 - 2.0}{3.3}$$

$$= \underline{0.90 \text{ s}}$$

7. June/2023/Paper_0625/22/No.9

A resultant force of 2.0 N acts on an object of mass 3.0 kg for 6.0 s.

What is the change in velocity of the object?

- A 0.25 m/s B 1.0 m/s (C) 4.0 m/s D 36 m/s

$v - u$ is change in velocity

$$a = \frac{F \cdot t}{m} = \frac{2.0}{3.0}$$

$$= 0.67 \text{ m/s}^2$$

$$\frac{v - u}{t} = a$$

$$v - u = a \times t$$

$$= 0.67 \times 6$$

$$= 4.02 \text{ m/s}$$

$$\approx 4.0 \text{ m/s}$$

8. June/2023/Paper_0625/23/No.9

A ball of mass 0.25 kg hits a wall at a speed of 16 m/s. It then rebounds back along its original path at a speed of 12 m/s.

What is the impulse experienced by the ball during its impact with the wall?

- A 1.0Ns B 3.0Ns C 4.0Ns **D 7.0Ns**

$$\begin{aligned} \text{Impulse} &= F \times t \\ F \times t &= mv - mu \\ u &= 16 \text{ m/s} \\ v &= -12 \text{ m/s} \end{aligned} \quad \begin{aligned} \text{Impulse} &= m(v - u) \\ &= 0.25(-12 - 16) \\ &= -7.0 \text{ Ns} \\ &= \underline{\underline{-7.0 \text{ Ns}}} \end{aligned}$$

9. June/2023/Paper_0625/42/No.2

A student catches a cricket ball. The speed of the ball immediately before it is caught is 18 m/s. The mass of the cricket ball is 160 g.

$$160 \text{ g} = 0.16 \text{ kg}$$

- (a) Calculate the kinetic energy stored in the cricket ball immediately before it is caught.

$$\begin{aligned} K.E &= \frac{1}{2} mv^2 \\ &= \frac{1}{2} \times 0.16 \times 18^2 \\ &= 25.92 \text{ J} \\ &\approx 26 \text{ J} \end{aligned}$$

kinetic energy = 26 J [3]

- (b) It takes 0.12 s to catch the ball and bring it to rest.

Calculate the average force exerted on the ball.

$$\begin{aligned} F &= \frac{mv - mu}{t} \\ &= \frac{m(v - u)}{t} \\ &= \frac{0.16 \times (0 - 18)}{0.12} \\ &= \underline{\underline{-24 \text{ N}}} \end{aligned}$$

average force = 24 N [2]

(c) As the student catches the ball, she moves her hands backwards.

Explain the effect of this action on the student's hands.

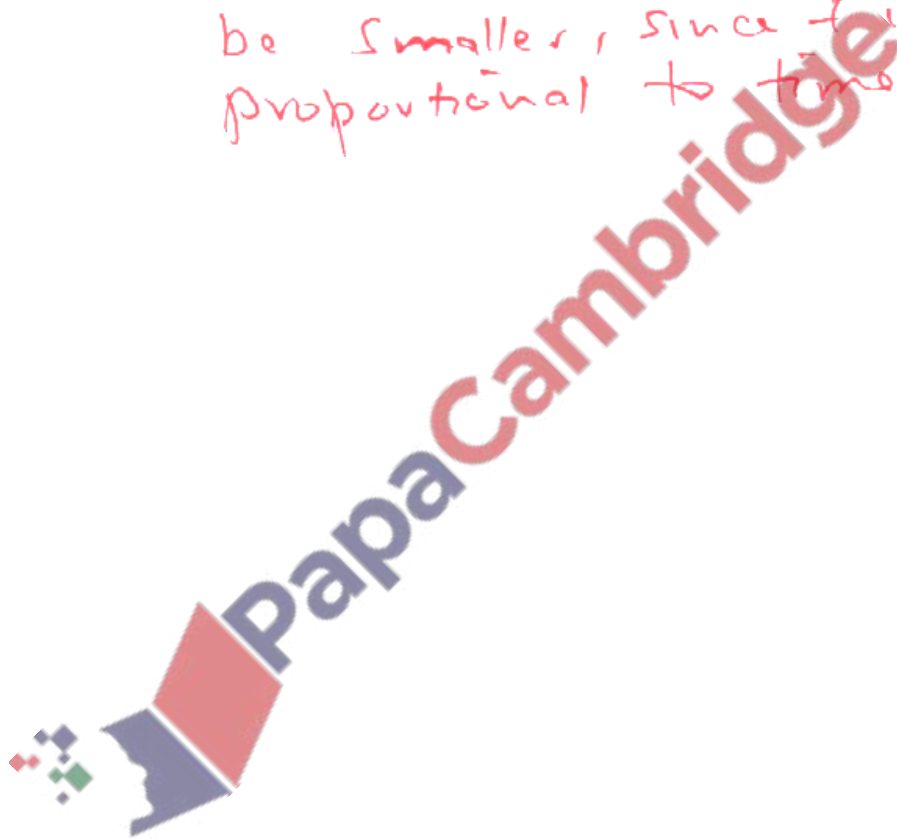
- This increases the time of impact
- So smaller force will act on the student's hand, thus it does not hurt as much.

[1]

[Total: 6]

$$F = \frac{mv - mu}{t}, \quad F \propto \frac{1}{t}$$

when t is large, the F will be smaller, since F is inversely proportional to time.



Energy, Work & Power

Question Paper 1

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Energy, Work & Power
Booklet	Question Paper 1

Time allowed: 21 minutes

Score: /17

Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

A lorry of mass 4000 kg is travelling at a speed of 4.0 m / s.

A car has a mass of 1000 kg. The kinetic energy of the car is equal to the kinetic energy of the lorry.

What is the speed of the car?

- A 2.0m/s B 4.0m/s C 8.0m/s D 16.0m/s

Question 2

A force acts on an object and causes the object to move a certain distance, in the same direction as the force.

Which row represents a situation in which the largest amount of work is done on the object by the force?

	force/N	distance moved/m
A	2.0	40.0
B	10.0	2.0
C	20.0	6.0
D	100.0	1.0

Question 3

A stone of mass m is held at rest in water. The stone is released and falls vertically a distance h . The stone reaches a speed v .

Some of the original energy of the stone is transferred to the water. As it falls, resistive forces cause the temperature of the water and stone to increase.

Which expression gives the work done against the resistive forces?

A $\frac{1}{2}mv^2$

B $mgh - \frac{1}{2}mv^2$

C mgh

D $mgh + \frac{1}{2}mv^2$

Question 4

Which energy resource does not derive its energy from the Sun?

- A geothermal
- B hydroelectric
- C oil
- D waves

Question 5

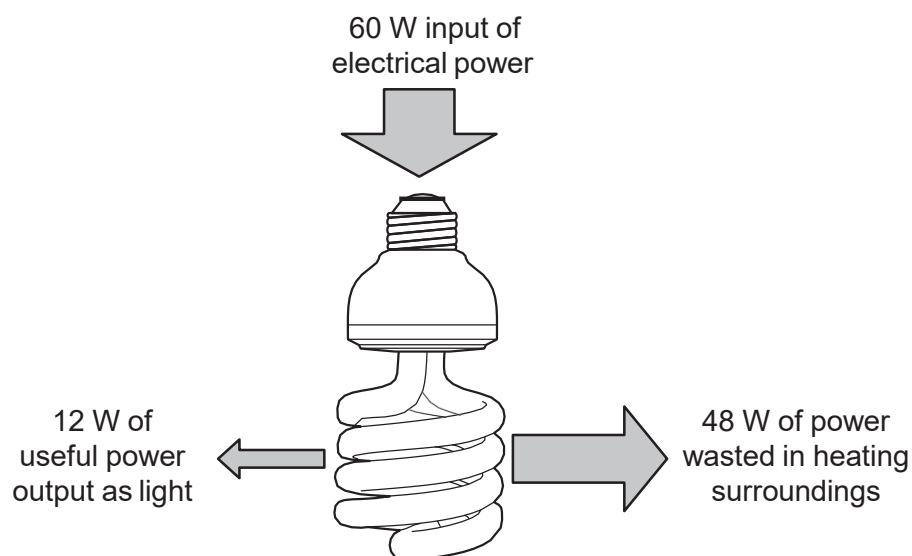
Some processes are more efficient than others.

Which expression gives the efficiency of a process?

- A $\frac{\text{total energy output}}{\text{total energy input}} \times 100\%$
- B $\frac{\text{useful energy output}}{\text{total energy input}} \times 100\%$
- C $\frac{\text{wasted energy output}}{\text{total energy input}} \times 100\%$
- D $\frac{\text{wasted energy output}}{\text{useful energy output}} \times 100\%$

Question 6

The diagram shows the energy used by a modern lamp.

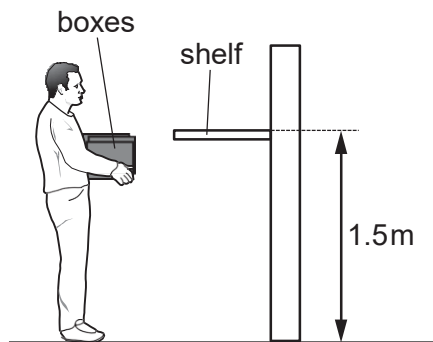


Which expression gives the efficiency of the lamp?

- A $\frac{12}{60} \times 100\%$ B $\frac{12}{48} \times 100\%$ C $\frac{48}{60} \times 100\%$ D $\frac{48}{12} \times 100\%$

Question 7

Three boxes each weigh 100 N. A man lifts all the boxes together from the ground on to a shelf that is 1.5 m above the ground. The man takes 2.0 s to do this.



How much useful power does the man produce to lift the boxes?

- A 75 W B 225 W C 300 W D 900 W

Question 8

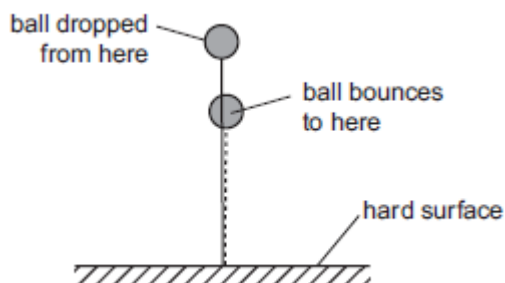
A car has a mass of 1000 kg and a momentum of 12000 kg m/s.

What is its kinetic energy?

- A. 6 kJ
- B. 12 kJ
- C. 72 kJ
- D. 144 kJ

Question 9

A ball is dropped on to a hard surface and bounces. It does not bounce all the way back to where it started, and so has not regained all of its original gravitational potential energy.



Which statement accounts for the loss of gravitational potential energy?

- A. Energy was destroyed as the ball hit the ground.
- B. Energy was destroyed as the ball travelled through the air.
- C. The chemical energy and elastic energy of the ball have increased.
- D. The internal (heat) energy of the ball and its surroundings has increased.

Question 10

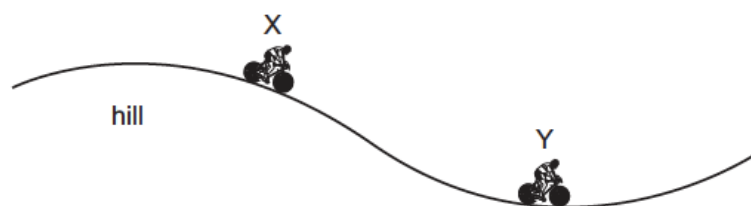
Which energy resource is used to generate electricity without using any moving parts?

- A geothermal
- B hydroelectric
- C nuclear
- D solar

Question 11

A cyclist travels down a hill from rest at point X, without pedalling.

The cyclist applies his brakes and the cycle stops at point Y.



Which energy changes have taken place between X and Y?

- A gravitational potential $\rightarrow$ kinetic $\rightarrow$ thermal (heat)
- B gravitational potential $\rightarrow$ thermal (heat) $\rightarrow$ kinetic
- C kinetic $\rightarrow$ gravitational potential $\rightarrow$ thermal (heat)
- D kinetic $\rightarrow$ thermal (heat) $\rightarrow$ gravitational potential

Question 12

Which source of energy involves the splitting of heavy atoms?

- A. chemical energy
- B. geothermal energy
- C. hydroelectric energy
- D. nuclear energy

Question 13

To calculate the power produced by a force, the size of the force must be known.

What else needs to be known to calculate the power?

	the distance the force moves the object	the time for which the force acts on the object
A	✓	✓
B	✓	X
C	X	✓
D	X	X

key

✓ = needed

X = not needed

Question 14

Electrical energy may be obtained from nuclear fission.

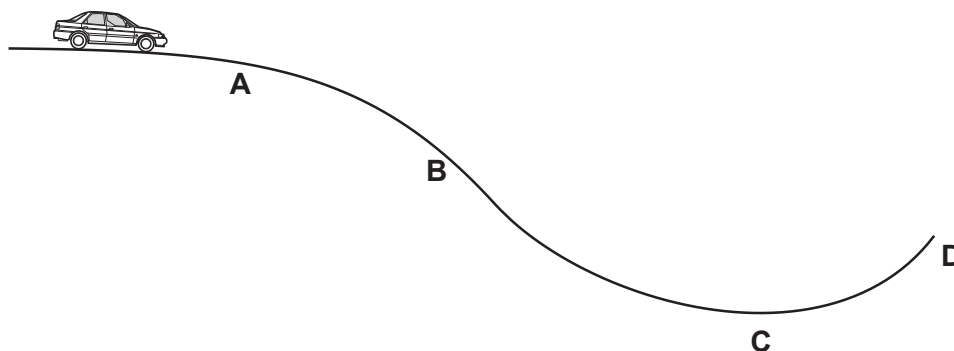
In which order is the energy transferred in this process?

- A nuclear fuel → generator → reactor and boiler → turbines
- B nuclear fuel → generator → turbines → reactor and boiler
- C nuclear fuel → reactor and boiler → generator → turbines
- D nuclear fuel → reactor and boiler → turbines → generator

Question 15

A car is stationary at the top of a hill with the engine switched off. The brakes are released and the car rolls down the hill.

At which labelled point does the car have the greatest kinetic energy? Ignore friction.



Question 16

Which row gives the energy change in a battery and the energy change in a solar cell?

	battery	solar cell
A	chemical to electrical	electrical to light
B	chemical to electrical	light to electrical
C	electrical to chemical	electrical to light
D	electrical to chemical	light to electrical

5

A helicopter takes off from the ground and rises vertically. It then hovers at a constant height above the ground.

Which sequence of energy changes takes place during the gain in height?

- A chemical $\rightarrow$ gravitational potential $\rightarrow$ kinetic
- B chemical $\rightarrow$ kinetic $\rightarrow$ gravitational potential
- C gravitational potential $\rightarrow$ chemical $\rightarrow$ kinetic
- D kinetic $\rightarrow$ chemical $\rightarrow$ gravitational potential

Energy, Work & Power

Question Paper 2

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Energy, Work & Power
Booklet	Question Paper 2

Time allowed: 24 minutes

Score: /19

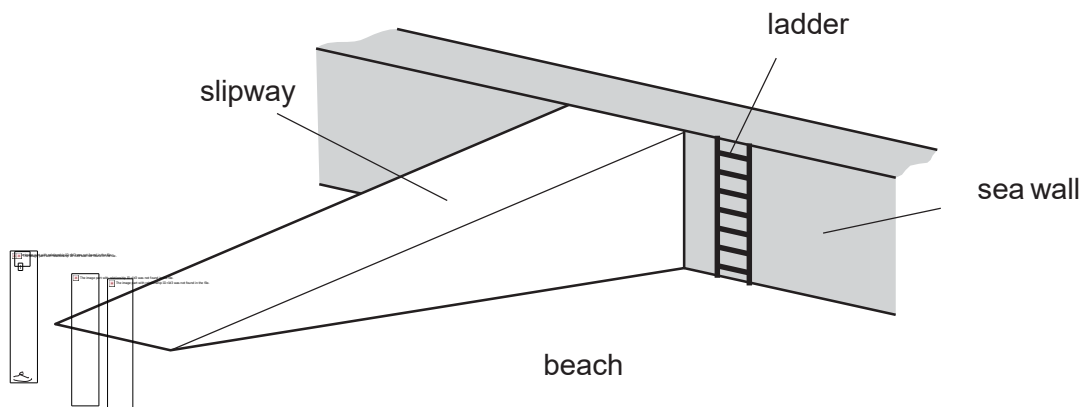
Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

Four people of equal weight on a beach use different routes to get to the top of a sea wall.



Which person produces the greatest average power?

person	route	time taken/s
A	runs across the beach, then climbs the ladder	8
B	walks across the beach, then climbs the ladder	16
C	runs up the slipway	5
D	walks up the slipway	10

Question 2

Which energy transfer takes place when a matchstick burns?

- A chemical to thermal
- B chemical to nuclear
- C nuclear to chemical
- D thermal to chemical

Question 3

Four cars are driven along a road.

The table shows the work done by the engine in each car and the time taken by each car.

Which engine produces the most power?

	work done by engine/J	time taken/s
A	50000	20
B	50000	40
C	100000	20
D	100000	40

Question 4

In a hydroelectric power station, one form of energy is stored in a lake or reservoir. This energy is then transferred in stages to another useful form, which is the output.

Which row gives the name of the stored energy and the name of the output energy?

	stored energy	output energy
A	electrical	thermal (heat)
B	electrical	kinetic
C	gravitational	electrical
D	kinetic	electrical

Question 5

A certain machine is very efficient.

What does this mean?

- A. It produces a large amount of power.
- B. It uses very little energy.
- C. It wastes very little energy.
- D. It works very quickly.

Question 6

The list contains three energy resources P, Q and R.

- P geothermal energy from hot rocks
- Q nuclear fission in reactors
- R sunlight on solar panels

Which of these resources are renewable?

- A P and Q only
- B P and R only
- C Q and R only
- D P, Q and R.

Question 7

Which movement will require the greatest amount of work to be done?

- A a force of 10 N moving an object a distance of 3.0 m
- B a force of 10 N moving an object a distance of 5.0 m
- C a force of 15 N moving an object a distance of 3.0 m
- D a force of 15 N moving an object a distance of 5.0 m

Question 8

When a bicycle lamp is switched on, what is the useful energy change **within the battery**?

- A. chemical energy to electrical energy
- B. electrical energy to chemical energy
- C. electrical energy to light energy
- D. light energy to chemical energy

Question 9

A student does some work by pulling a suitcase along a corridor.

She now pulls a second suitcase along the corridor.

Which row indicates that the student is now doing twice as much work?

	the force used to pull suitcase	the distance the suitcase is pulled
A	is doubled	is doubled
B	is doubled	is halved
C	stays the same	is doubled
D	stays the same	is halved

Question 10

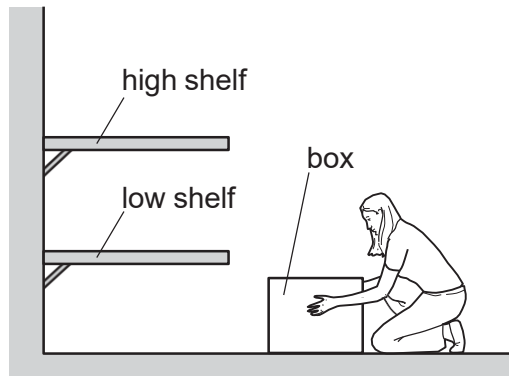
Some energy sources are reliably available at all times, and some are not.

Which row shows three sources all in their correct columns?

	available at all times	not available at all times
A	geothermal	nuclear fission, solar
B	geothermal, nuclear fission	solar
C	solar, nuclear fission	geothermal
D	solar	nuclear fission, geothermal

Question 11

A woman in a factory has to lift a box on to a shelf.



Which action involves the woman in doing the **least** amount of work?

- A. lifting the box quickly to the high shelf
- B. lifting the box slowly to the high shelf
- C. lifting the box to the low shelf first then lifting it to the high shelf
- D. lifting the box to the low shelf instead of to the high shelf

Question 12

An aeroplane is landing. As it descends towards the runway, its speed reduces.

What are the energy changes that take place during the descent?

- A kinetic + gravitational $\rightarrow$ thermal (heat)
- B kinetic $\rightarrow$ gravitational + thermal (heat)
- C kinetic + thermal (heat) $\rightarrow$ gravitational
- D thermal (heat) $\rightarrow$ kinetic + gravitational

Question 13

Energy from uranium is transferred to electrical energy in a nuclear power station.

What is the correct order of the stages of this process?

- A boiler → generator → reactor → turbine
- B generator → boiler → turbine → reactor
- C reactor → boiler → turbine → generator
- D reactor → turbine → boiler → generator

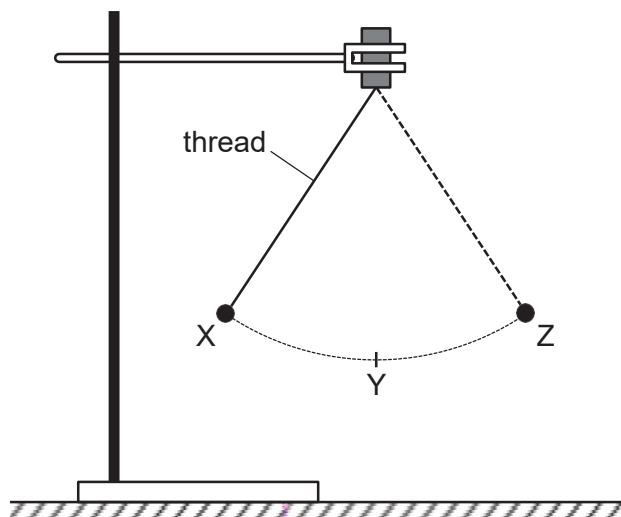
Question 14

In which pair of energy sources are both sources renewable?

- A. oil and coal
- B. oil and tidal
- C. tidal and geothermal
- D. tidal and nuclear fission

Question 15

An object on a thread is swinging between X and Z, as shown in the diagram. It is momentarily at rest at X and at Z.



An incomplete word equation about the energy of the object is shown below.

gravitational potential energy = kinetic energy + energy + energy losses at X
at Y at Y

Which form of energy is needed to complete the word equation?

- A. chemical
- B. gravitational potential
- C. internal
- D. Strain.

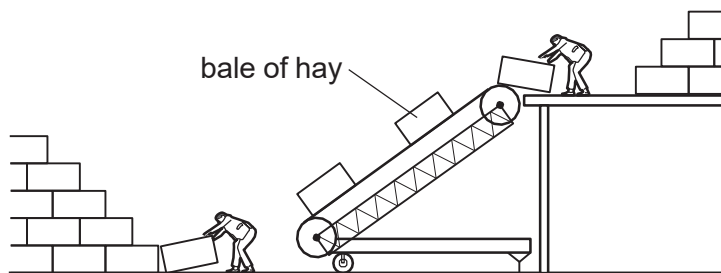
Question 16

Which row gives an example of the stated form of energy?

	form of energy	example
A	gravitational	the energy due to the movement of a train along a level track
B	internal	the energy due to the flow of cathode rays in a cathode-ray tube
C	kinetic	the energy due to the position of a swimmer standing on a high diving board
D	strain	the energy due to the compression of springs in a car seat

Question 17

Two farmers use an electrically powered elevator to lift bales of hay. All the bales of hay have the same mass.



As sunset approaches, they increase the speed of the motor so that more bales are lifted up in a given time.

How does this affect the work done in lifting each bale and the useful output power of the motor?

	work done in lifting each bale	useful output power of the motor
A	increases	decreases
B	increases	increases
C	no change	decreases
D	no change	increases

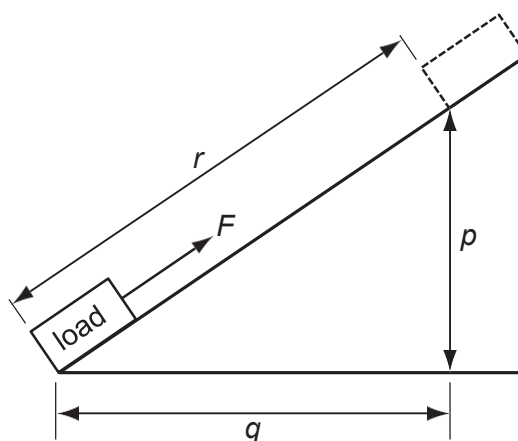
Question 18

Which energy resource is used to generate electricity by first boiling water?

- A hydroelectric
- B nuclear fission
- C tides
- D waves

Question 19

A force F moves a load from the bottom of a slope to the top.



The work done by the force depends on the size of the force, and on a distance.

What is this distance?

- A p B q C r D $p + q$

1. Nov/2023/Paper_0625/11/No.9

Which energy resource is non-renewable?

- A** geothermal
- B** natural gas
- C** solar
- D** wind

2. Nov/2023/Paper_0625/12/No.9

In a small power station, biofuel is used to generate electricity.

Which energy store is reduced by this process?

- A** chemical
- B** kinetic
- C** nuclear
- D** thermal

3. Nov/2023/Paper_0625/13/No.9

Electrical power is generated from different resources. Some of these resources are listed.

- chemical energy stored in biofuels
- chemical energy stored in fossil fuels
- energy stored in tides
- geothermal resources
- hydroelectric resources
- light from the Sun
- nuclear fuel

How many of the resources listed are classified as renewable?

- A** 3 **B** 4 **C** 5 **D** 6

Fig. 6.1 shows four wind turbines.

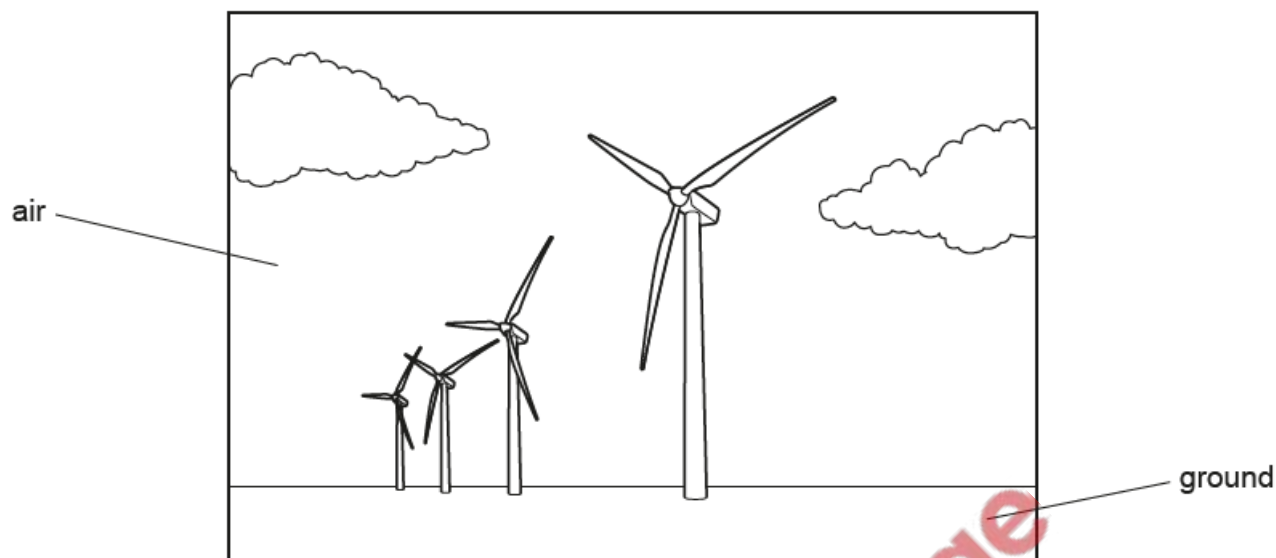


Fig. 6.1

(a) Describe how a wind turbine generates electrical power.

.....

.....

..... [3]

(c) For transmission, the output voltage is increased to 132 kV.

State **two** advantages of transmitting electrical power at high voltage.

1

2 [2]

Many methods of generating electrical power involve the use of water.

- (a) Describe **one** method of generating electrical power from energy stored in water.

.....

.....

.....

.....

..... [3]

- (b) For the method you chose in (a), state **one** advantage and **one** disadvantage of generating electricity this way.

advantage

.....

disadvantage

..... [2]

- (c) State **two** methods of generating electrical power for which the main source of energy is **not** the Sun.

1

2

[2]

[Total: 7]



6. Nov/2023/Paper_0625/43/No.5

- (c) (i) Describe how thermal energy from nuclear reactions is used to generate electricity in a power station.

.....

.....

.....

..... [3]

- (ii) State **one** advantage and **one** disadvantage of using nuclear fuels in a power station instead of using fossil fuels.

advantage

.....

disadvantage

..... [2]

7. June/2023/Paper_0625/23/No.11

Research is being carried out to produce electrical energy from the fusion of hydrogen nuclei.

Which row shows two of the problems in designing a fusion reactor?

	temperature needed	why obtaining a high density of hydrogen nuclei is difficult
A	very low	the nuclei are negatively charged and repel each other
B	very low	the nuclei are positively charged and repel each other
C	very high	the nuclei are negatively charged and repel each other
D	very high	the nuclei are positively charged and repel each other

Electricity is distributed from wind turbines to homes and industry.

- (a) Statements A–F describe the main stages in the transfer of energy from the Sun to electrical energy in a wind turbine generator.

The statements A–F are **not** in the correct order.

- A Air moves from regions of high pressure to regions of low pressure.
- B The turbine blades turn a generator.
- C Energy from the Sun heats the atmosphere unevenly.
- D Uneven heating of the atmosphere produces regions of different atmospheric pressure.
- E The generator produces electrical energy.
- F Moving air turns the turbine blades.

Complete the flow chart to describe how a wind turbine uses energy from the Sun to generate electrical energy. Insert the missing letters in the empty boxes.



[3]

- (b) State **two** disadvantages, apart from cost, of using wind turbines to produce electrical energy for homes and industry.

1

2

[2]

[Total: 5]

Fig. 2.1 shows an engineer working with wind turbines.



Fig. 2.1

(a) Complete the sentences describing how electrical power is generated by energy in the wind.

(i) The source of the wind energy is [1]

(ii) When the blades turn, electrical power is generated in the [1]

(b) Describe **two** advantages, apart from cost, of generating electrical power by using wind turbines compared with using a coal-fired power station.

1

.....

2

.....

[2]

[Total: 4]

(b) State **two** energy resources for which the Sun is **not** the main source.

1

2

[2]

(c) State and explain whether each of the following methods of electrical power generation is renewable.

(i) power generation in a nuclear power station

statement

explanation

.....

[2]

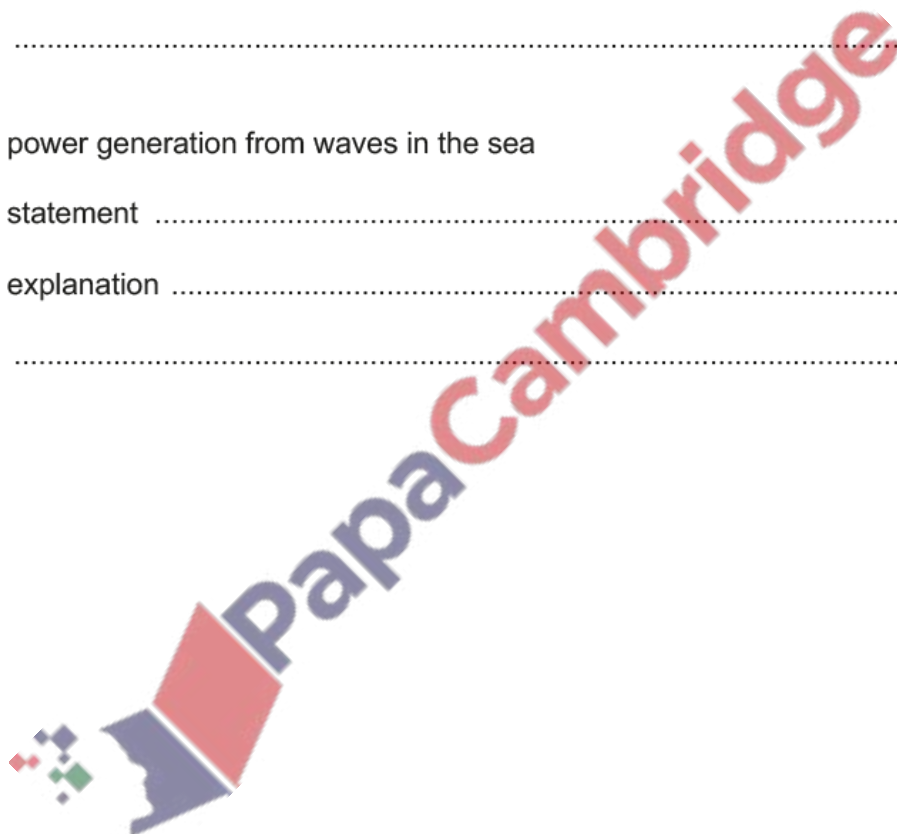
(ii) power generation from waves in the sea

statement

explanation

.....

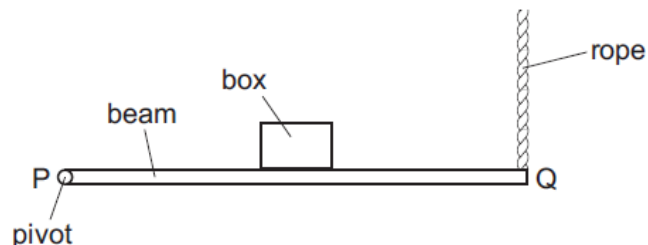
[2]



1. Nov/2023/Paper_0625/11/No.7

The diagram shows a wooden beam PQ which is attached to a wall by a pivot at P and kept in a horizontal position by a vertical rope attached at Q.

A box has been placed on the beam.

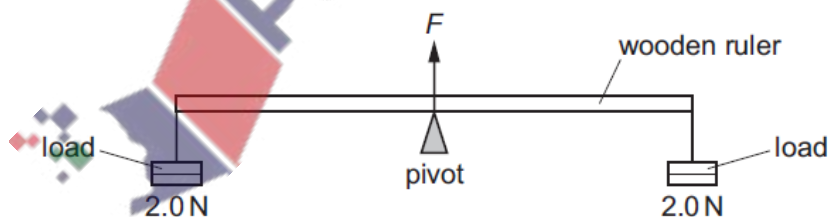


Which changes **must** reduce the tension in the rope at Q?

- A Decrease the mass of the box and move it towards P.
- B Decrease the mass of the box and move it towards Q.
- C Increase the mass of the box and move it towards P.
- D Increase the mass of the box and move it towards Q.

2. Nov/2023/Paper_0625/12/No.7

A uniform wooden ruler is pivoted at its centre. A load of 2.0 N is suspended from each end of the ruler.



The pivot exerts an upward force F on the ruler.

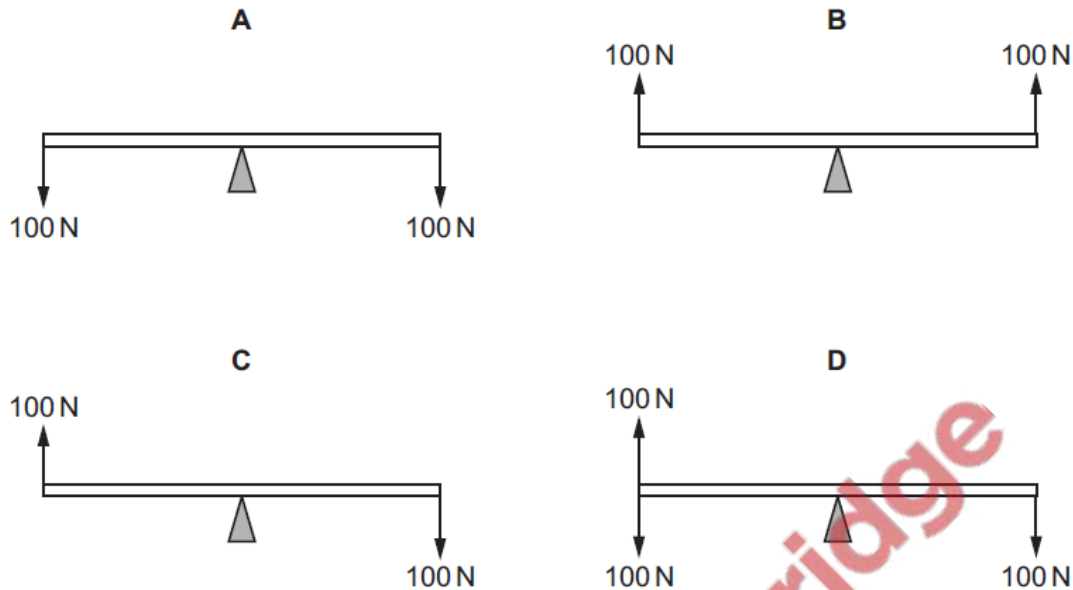
What is F equal to?

- A 2.0 N
- B the weight of the ruler
- C 4.0 N
- D 4.0 N plus the weight of the ruler

3. Nov/2023/Paper_0625/13/No.6

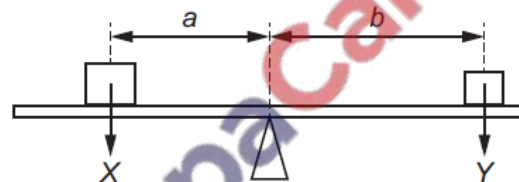
A uniform rod rests on a pivot at its centre. The rod is not attached to the pivot. Forces are then applied to the rod in four different ways, as shown. The weight of the rod can be ignored.

Which diagram shows the rod in equilibrium?



4. Nov/2023/Paper_0625/13/No.7

The diagram shows a beam balanced on a pivot. Two forces, X and Y , are acting on the beam.



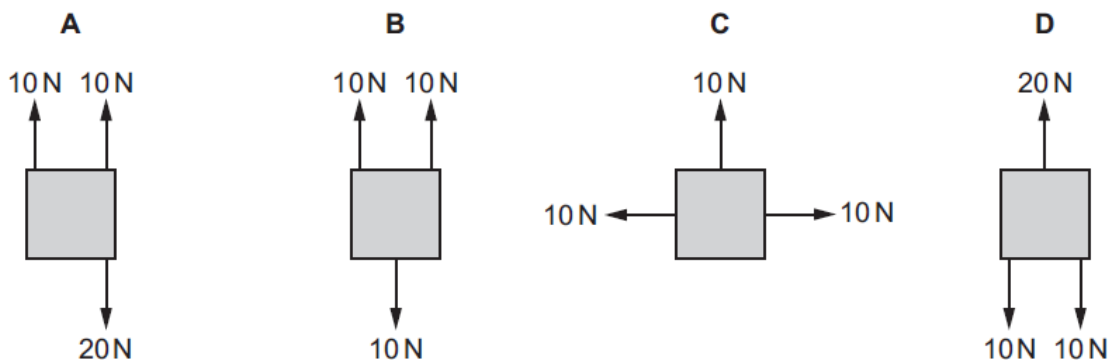
Which calculation gives the moment of the force Y about the pivot?

- A $Y \times (a + b)$ B $\frac{Y}{(a + b)}$ C $Y \times b$ D $\frac{Y}{b}$

5. Nov/2023/Paper_0625/22/No.6

The diagrams show four identical objects. Each object is acted on by only the forces shown.

Which diagram shows an object in equilibrium?



6. Nov/2023/Paper_0625/31/No.2(c)

- (c) The student balances a beam on a pivot. On the beam, he positions the cylinder and a block so that the beam remains balanced. The arrangement is shown in Fig. 2.2.

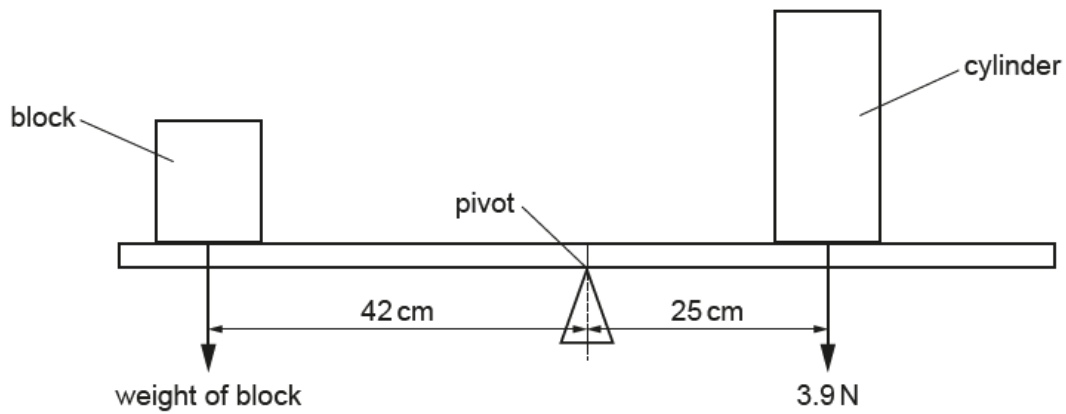


Fig. 2.2 (not to scale)

Calculate the weight of the block.

weight of block = N [4]

A platform rests on a pivot as shown in Fig. 3.1.

A diver sits at a distance of 1.8 m from the pivot. The weight of the diver is 1100 N.

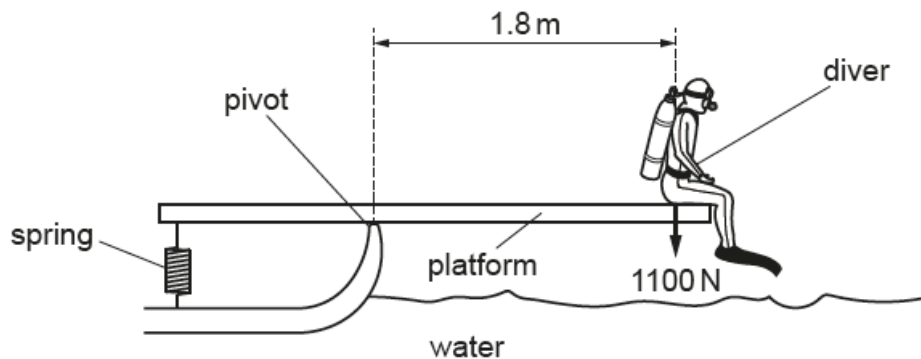


Fig. 3.1 (not to scale)

(a) Using the information in Fig. 3.1, calculate the moment of the diver about the pivot.

moment of diver = Nm [3]

(b) (i) Fig. 3.2 represents the platform without the diver.

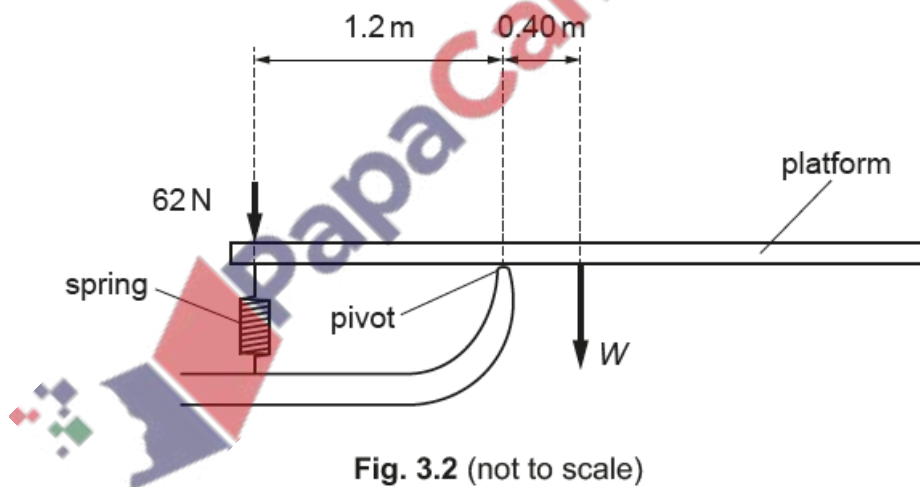


Fig. 3.2 (not to scale)

The moment of the weight W of the platform is balanced by the moment of the spring. The spring exerts a downward force of 62 N.

Using the information in Fig. 3.2, calculate the weight W of the platform.

$W =$ N [3]

A radio transmitter is a very tall, thin cylinder. It is prevented from falling over by wires which have one end fixed to the transmitter and the other end fixed in the ground. The ends of the wires in the ground are a long distance from the transmitter.

Fig. 4.1 shows the transmitter and two of the wires.

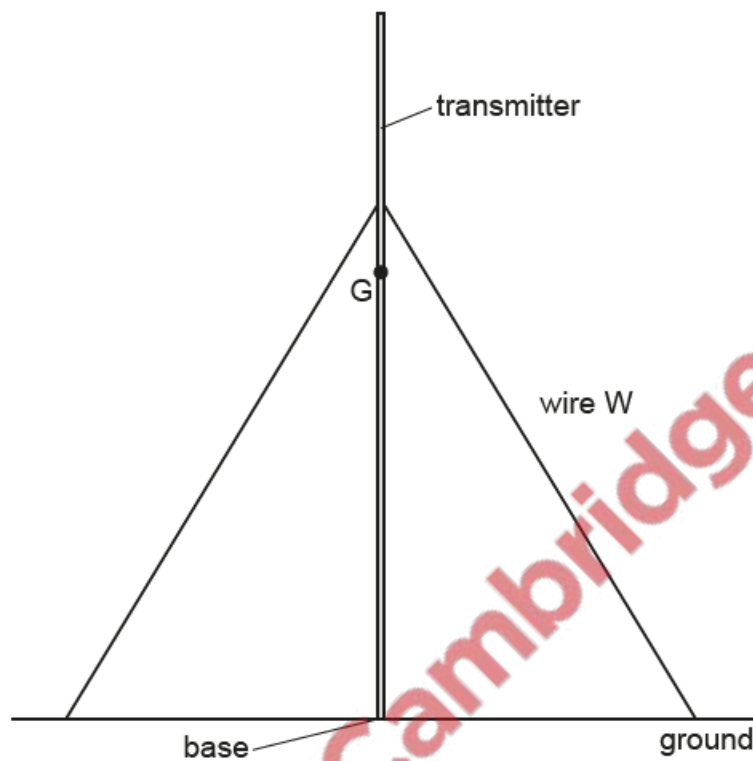


Fig. 4.1

(a) The centre of gravity G is shown on Fig. 4.1.

(i) State what is meant by centre of gravity.

..... [1]

(ii) Explain why the radio transmitter without the wires is a very unstable structure.

..... [1]

(b) Wire W is under tension and it exerts a force T on the transmitter.

(i) On Fig. 4.1, mark an arrow to show the force T exerted by wire W on the transmitter. [1]

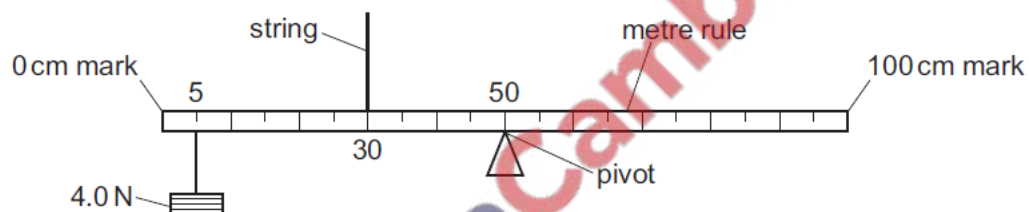
(ii) The force T produces a moment on the transmitter about its base.

Describe how the moment produced by T is calculated and indicate on Fig. 4.1 what is meant by any other terms in the description.

.....
..... [3]

9. June/2023/Paper_0625/11/No.8

The diagram shows a uniform metre rule. The rule is pivoted at its mid-point. A weight of 4.0 N is suspended from the rule at the 5 cm mark. The rule is held by a string at the 30 cm mark. The rule is in equilibrium.

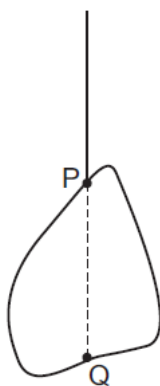


What is the upward force that the string exerts on the rule?

- A 0.67 N B 4.0 N C 6.0 N D 9.0 N

10. June/2023/Paper_0625/11/No.9

An irregularly shaped metal plate is freely suspended from a point P and is in equilibrium, as shown. Point Q is vertically below P.



Which statement about the position of the centre of gravity of the metal plate is correct?

- A It is at P.
- B It is at Q.
- C It is halfway between P and Q.
- D Further investigation is needed to determine its position.

11. June/2023/Paper_0625/12/No.7

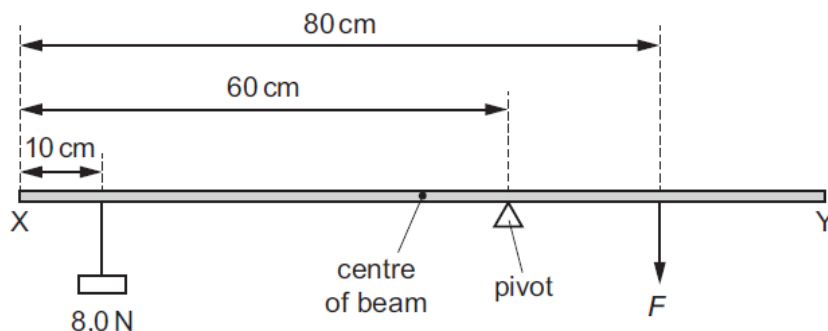
A uniform metre rule is pivoted in equilibrium at the 50 cm mark. A mass of 25 g is placed at the 30 cm mark on the rule.

What is the smallest mass that can be placed on the rule to restore equilibrium?

- A 5 g
- B 10 g
- C 15 g
- D 25 g

12. June/2023/Paper_0625/12/No.8

A uniform beam XY is 100 cm long and weighs 4.0 N.



The beam rests on a pivot 60 cm from end X.

A load of 8.0 N hangs from the beam 10 cm from end X.

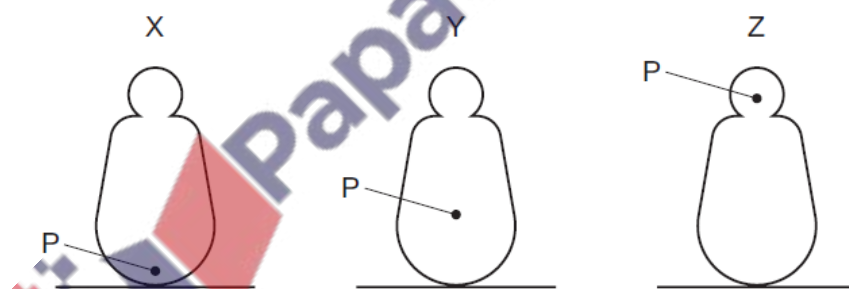
The beam is kept balanced by a force F acting on the beam 80 cm from end X.

What is the magnitude of force F ?

- A 8.0 N B 18 N C 22 N D 44 N

13. June/2023/Paper_0625/12/No.9

Three children's toys, X, Y and Z, are the same size and shape. They have weights at different positions inside so that the position of the centre of gravity of each toy is different. Each toy's centre of gravity is marked P.



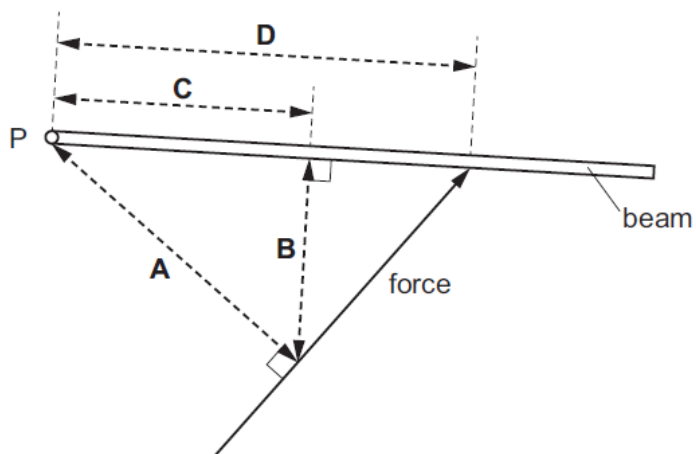
Which toy is the most stable and which toy is the least stable when balanced in the positions shown?

	most stable	least stable
A	X	Y
B	X	Z
C	Y	X
D	Y	Z

14. June/2023/Paper_0625/13/No.7

A beam is pivoted at P. A force is applied to the beam.

Which distance is multiplied by the force to give the moment of the force about P?



15. June/2023/Paper_0625/13/No.8

The diagram shows an unbalanced rod. Two loads X and Y can be moved along the rod.



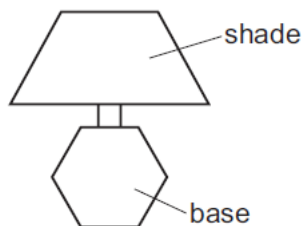
The rod turns in a clockwise direction, as shown.

Which action could make the rod balance?

- A moving X to the left
- B moving X to the right
- C moving Y to the right
- D moving the pivot to the left

16. June/2023/Paper_0625/13/No.9

It is important that an electric table lamp with a lamp shade does not get knocked over easily.



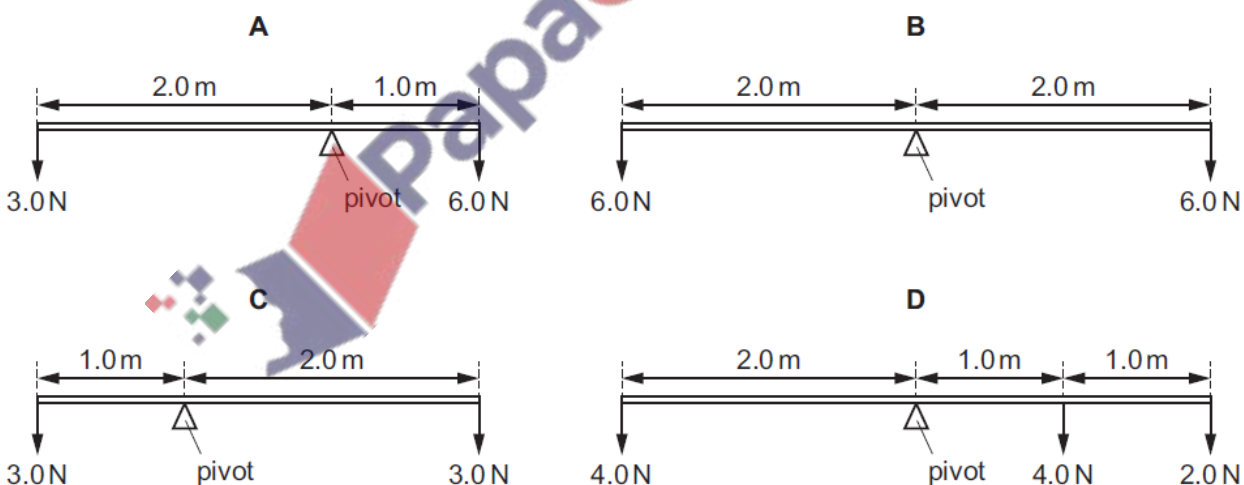
Which statement is correct?

- A The centre of mass must be as low as possible.
- B The lamp must have a shade which is heavier than the base.
- C The lamp must have a narrow base.
- D The lamp shade must be wide.

17. June/2023/Paper_0625/21/No.8

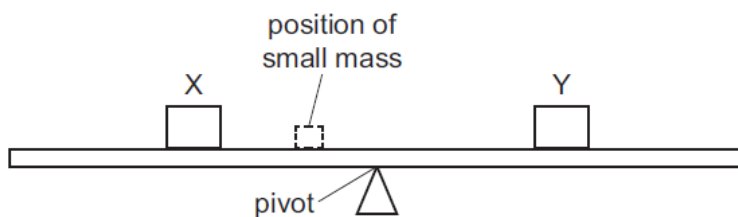
The diagrams show four beams, each of negligible weight and freely pivoted.

Which beam is **not** in equilibrium?



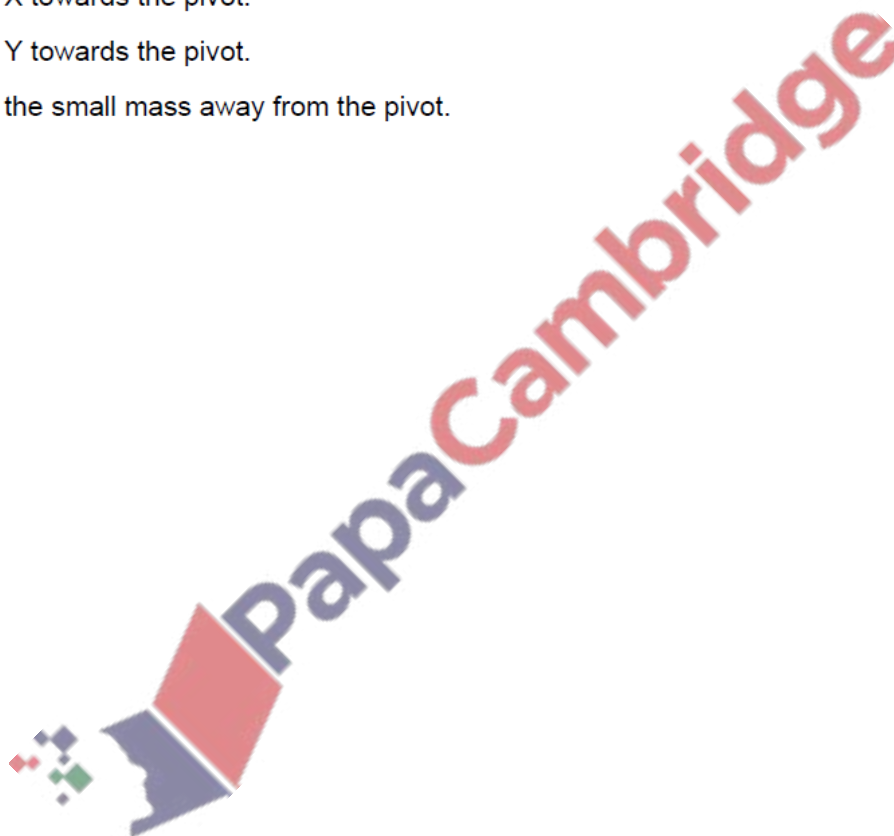
A uniform beam is pivoted at the centre and two identical masses, X and Y, are placed so that the beam balances.

A smaller mass is then added at the position shown.

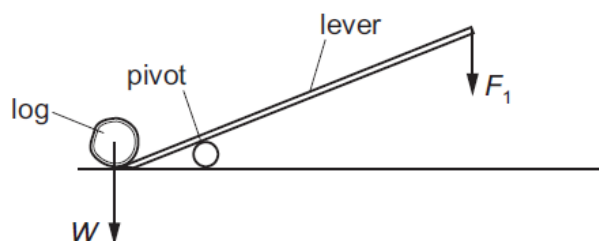


How can the masses be positioned so the beam balances again?

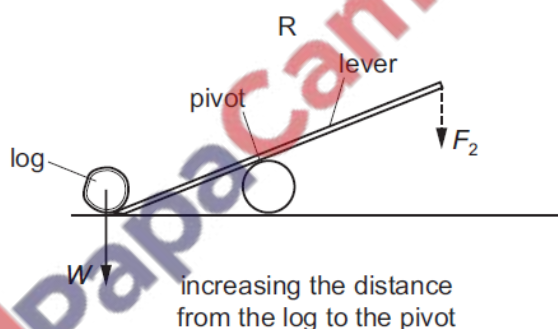
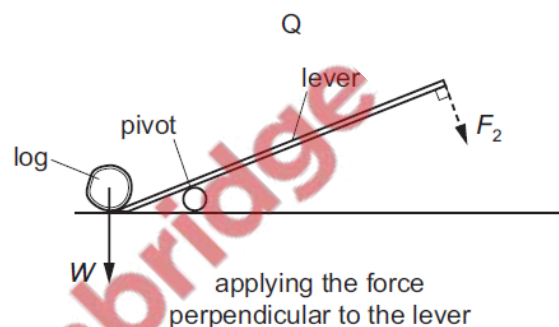
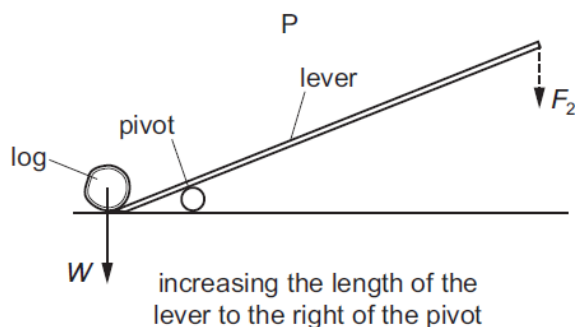
- A Move X away from the pivot.
- B Move X towards the pivot.
- C Move Y towards the pivot.
- D Move the small mass away from the pivot.



The diagram shows the minimum force F_1 acting vertically on a lever required to lift a heavy log of weight W .



The log needs to be lifted by a smaller force than F_1 . The diagrams show the changes tried. Each diagram has only **one** change from the original diagram. In each case, F_2 is the minimum force required to lift the log.



In which situations will F_2 be smaller than F_1 ?

- A** P, Q and R **B** P and Q only **C** P only **D** Q and R only

Fig. 2.1 shows a concrete beam resting on the ground.

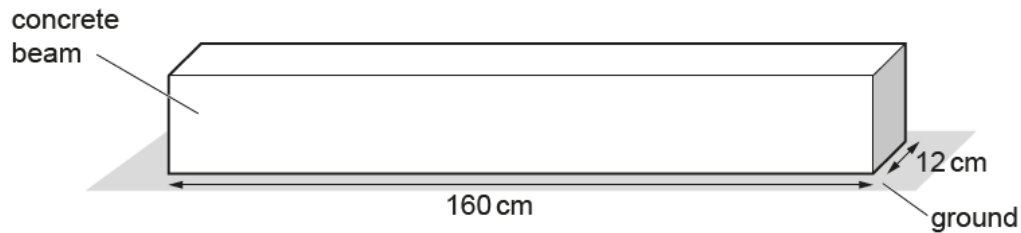


Fig. 2.1 (not to scale)

(b) A builder starts to raise one end of the beam.

He uses a force of 1030 N at a perpendicular distance of 120 cm from the pivot. Fig. 2.2 shows the arrangement.

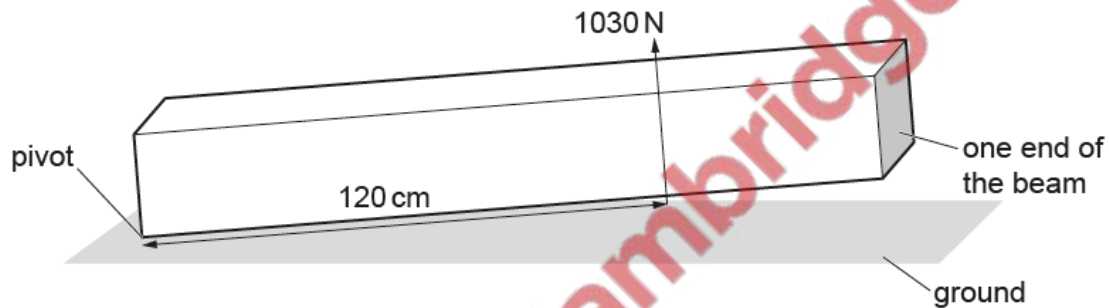


Fig. 2.2 (not to scale)

Calculate the moment of the 1030 N force about the pivot.

moment = N cm [3]

(c) Describe how the builder can use a smaller force to lift the beam.

..... [1]

(d) The builder positions the beam as shown in Fig. 2.3.

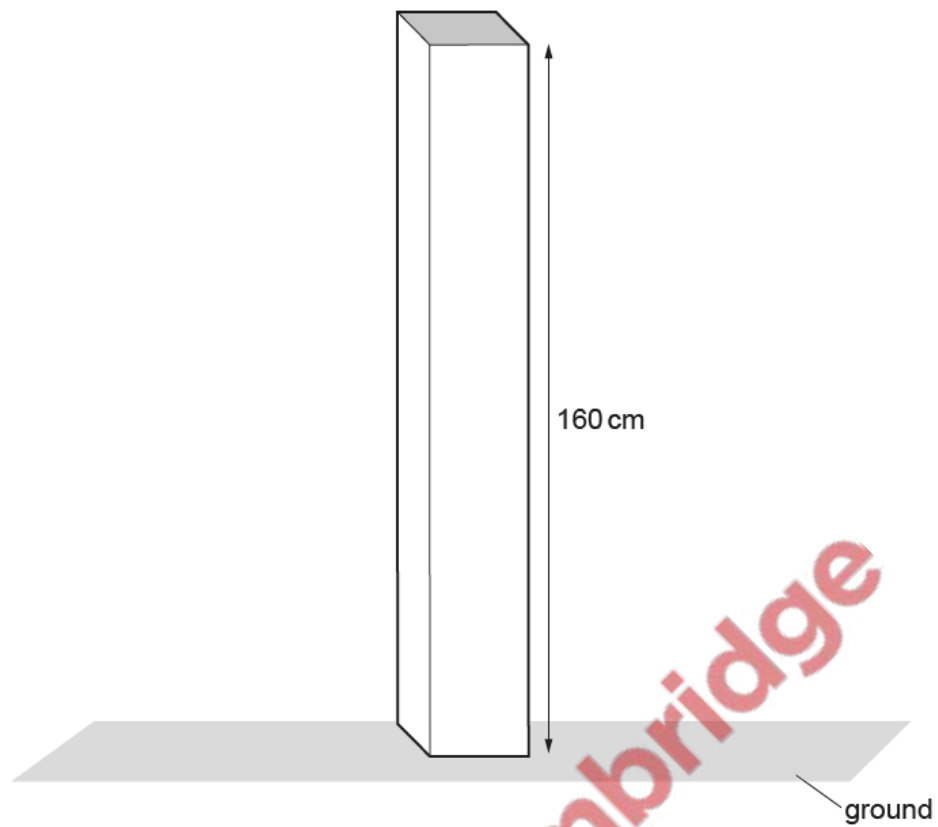


Fig. 2.3 (not to scale)

State why the beam shown in Fig. 2.3 is less stable than the beam shown in Fig. 2.1.

..... [1]

A student balances a beam on a pivot. They then balance block A and block B on the beam, as shown in Fig. 3.1.

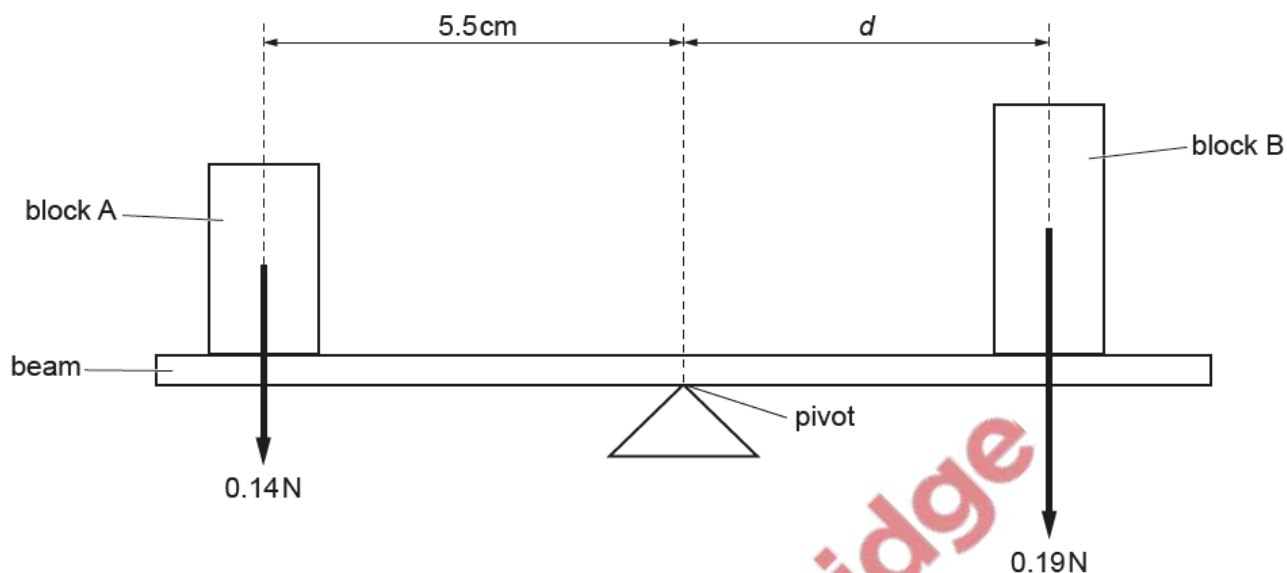


Fig. 3.1 (not to scale)

- (a) (i) The weight of block A is 0.14 N.

Show that the moment of block A about the pivot is approximately 0.8 N cm.

- (ii) The weight of block B is 0.19 N.

Calculate the distance d between the pivot and the centre of block B.

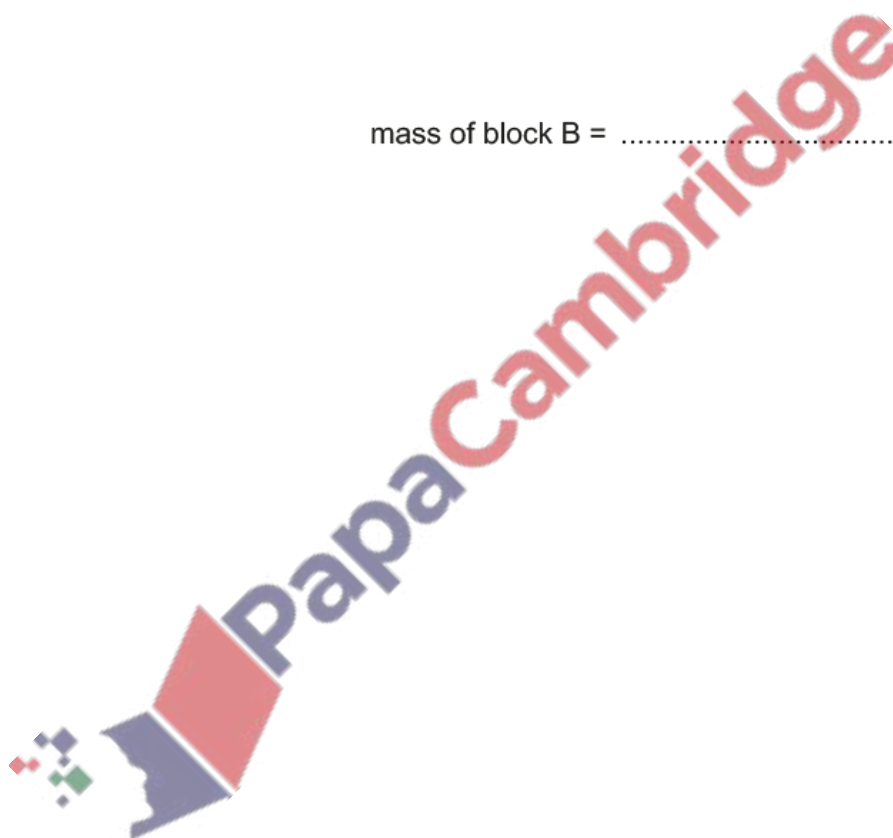
distance $d = \dots\dots\dots$ cm [3]

- (b) The weight of block B is 0.19 N.

Calculate the mass of block B.

mass of block B = $\dots\dots\dots$ kg [3]

[Total: 9]



Pressure

Question Paper 1

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Pressure
Booklet	Question Paper 1

Time allowed: 21 minutes

Score: /17

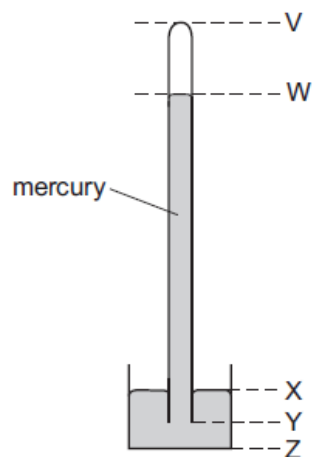
Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

The diagram shows a simple mercury barometer.



The atmospheric pressure increases.

Which distance increases?

A VW

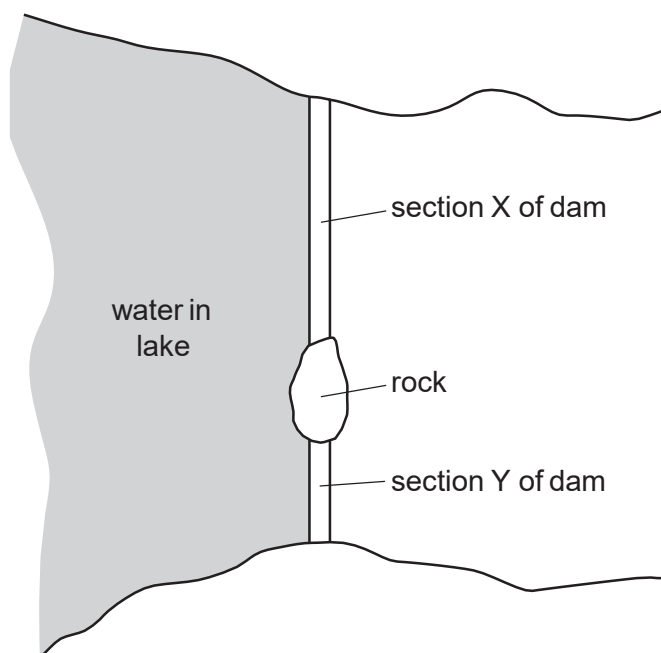
B WY

C XY

D XZ

Question 2

A dam across a lake is divided into two sections by a rock. Section X is longer than section Y but the two sections are otherwise identical. The water in the lake by the dam is the same depth everywhere. The diagram shows a view from above of the lake and the dam.



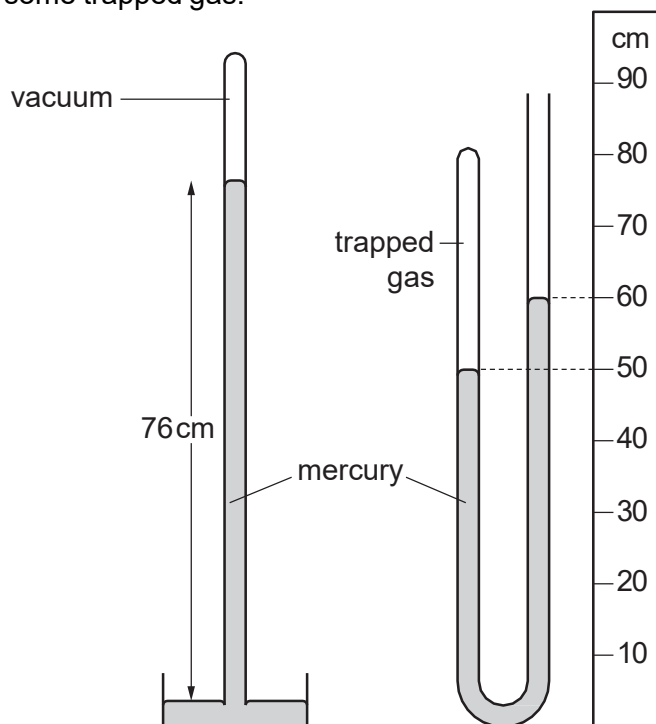
The water creates a total force on each section of the dam and an average pressure on each section of the dam.

Which statement is correct?

- A. The average pressure on X equals the average pressure on Y.
- B. The average pressure on X is less than the average pressure on Y.
- C. The total force on X equals the total force on Y.
- D. The total force on X is less than the total force on Y.

Question 3

The diagram shows a simple mercury barometer alongside a mercury manometer. The manometer contains some trapped gas.



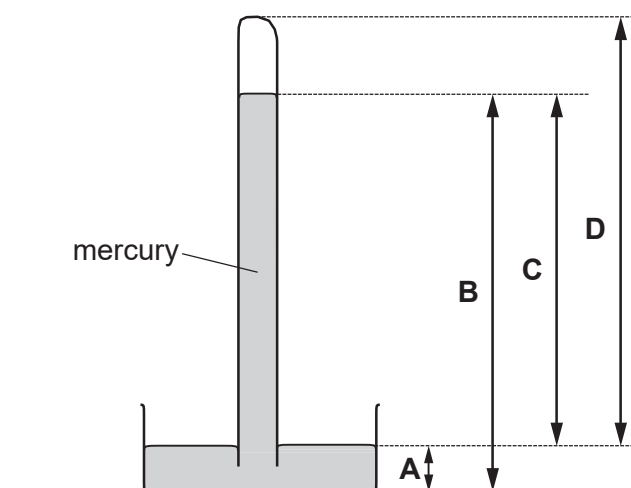
What is the pressure of the trapped gas?

- A 10 cm of mercury
- B 50 cm of mercury
- C 66 cm of mercury
- D 86 cm of mercury

Question 4

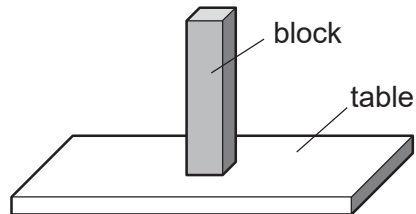
The diagram shows a simple mercury barometer.

Which height is used as a measurement of atmospheric pressure?

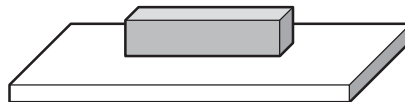


Question 5

A block with flat, rectangular sides rests on a table.



The block is now turned so that it rests with its largest side on the table.

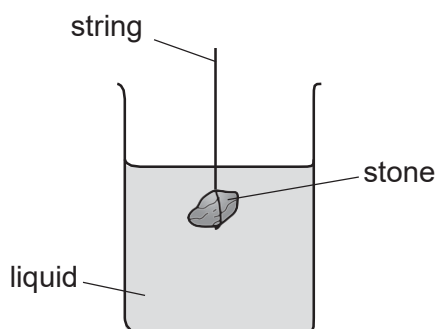


How has this change affected the force and the pressure exerted by the block on the table?

	force	pressure
A	decreased	decreased
B	decreased	unchanged
C	unchanged	decreased
D	unchanged	unchanged

Question 6

The diagram shows a stone suspended under the surface of a liquid from a string. The stone experiences a pressure caused by the liquid.

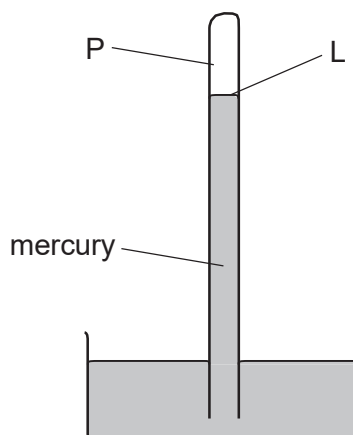


What would increase the pressure on the stone?

- A. decreasing the surface area of the stone
- B. increasing the mass of the stone
- C. lowering the stone deeper into the liquid
- D. using a liquid with a lower density

Question 7

The diagram shows a simple mercury barometer, used to measure atmospheric pressure.



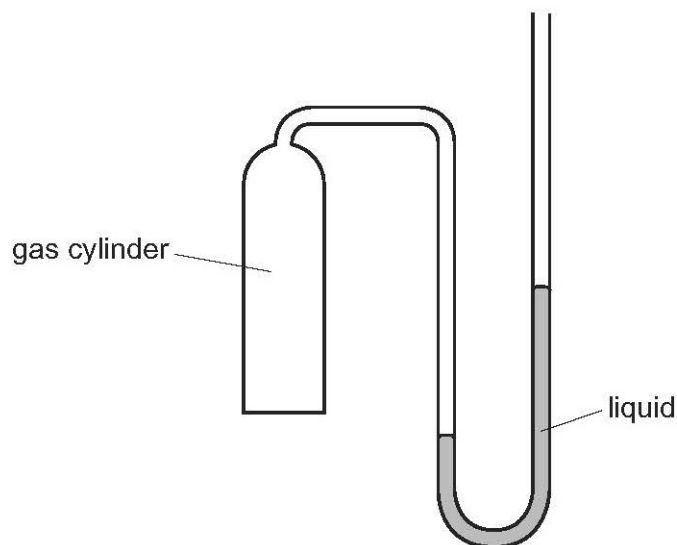
Atmospheric pressure decreases.

Which row states what happens to the pressure at point P and what happens to the level L?

	pressure at P	level L
A	decreases	falls
B	decreases	rises
C	stays the same	falls
D	stays the same	rises

Question 8

The diagram shows a manometer with one side connected to a gas cylinder and the other side open to the atmosphere.

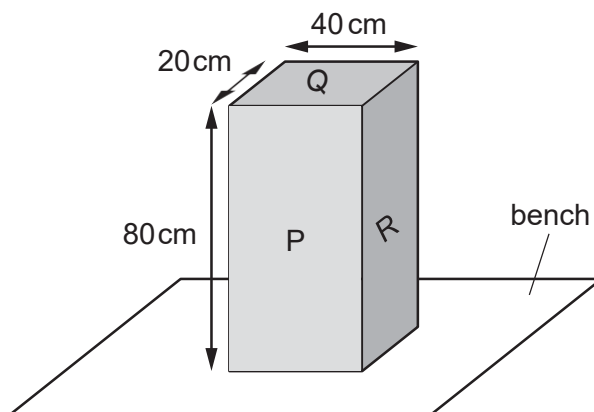


Which conclusion can be made using only the information from liquid levels in the manometer?

- A. The density of the gas is less than the density of air.
- B. The density of the gas is greater than the density of air.
- C. The pressure of the gas is less than atmospheric pressure.
- D. The pressure of the gas is greater than atmospheric pressure.

Question 9

The diagram shows a solid block resting on a bench. The dimensions of the block are shown.



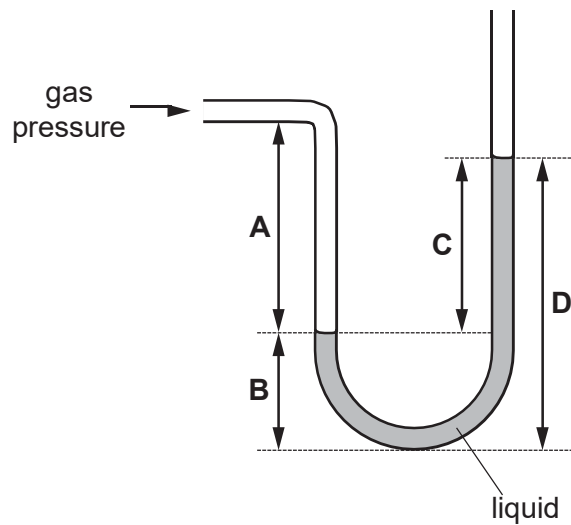
On which labelled surface should the block rest to produce the smallest pressure on the bench?

- A. P
- B. Q
- C. R
- D. any of P, Q or R

Question 10

The diagram shows a manometer containing a liquid. The manometer is used to find the difference between the pressure of a gas and atmospheric pressure.

Which distance represents this pressure difference?



Question 11

Four physics teachers investigate pressure. They wear identical clothes and lie on different beds of nails.

The table gives the weight of each teacher and the total area of contact between the teacher and the nails.

Which teacher experiences the least pressure from the nails?

	weight of teacher/N	total area of contact/cm ²
A	700	13
B	800	20
C	900	14
D	1000	21

Question 12

The table gives four forces and the surface area on which each force acts.

Which row gives the largest pressure on the surface?

	force/N	area/m ²
A	20	2
B	40	2
C	20	4
D	40	4

Question 13

What does a barometer measure?

- A the current in a circuit
- B the density of a liquid
- C the pressure of air
- D the temperature of an object

Question 14

A man is in contact with the floor.

In which of these situations does he produce the least pressure on the floor?

- A. kneeling
- B. lying flat on his back
- C. standing on both feet
- D. standing on one foot

Question 15

The water in a lake is at 5°C . A diver measures the pressure of the water at two different depths in the lake. He repeats the measurements on a different day when the water is at 15°C .

The density of the water decreases when its temperature increases.

Which combination of depth and temperature produces the greatest water pressure?

	depth / m	temperature / $^{\circ}\text{C}$
A	10	5
B	10	15
C	20	5
D	20	15

Question 16

A vehicle sinks into soft ground.

The vehicle is changed so that it does not sink as far.

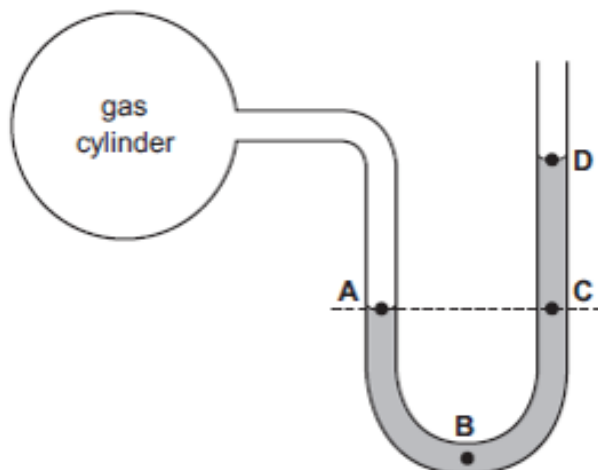
Which change is made?

- A. a lower centre of mass
- B. a more powerful engine
- C. wheels that are further apart
- D. wider tyres

Question 17

A manometer is used to measure the pressure of a gas trapped in a cylinder.

At which labelled point on the diagram is the pressure greatest?



Pressure

Question Paper 2

Level	IGCSE
Subject	Physics (0625/0972)
Exam Board	Cambridge International Examinations (CIE)
Topic	General Physics
Sub-Topic	Pressure
Booklet	Question Paper 2

Time allowed: 21 minutes

Score: /17

Percentage: /100

Grade Boundaries:

9	8	7	6	5	4	3	2	1
>85%	75%	68%	60%	55%	50%	43%	35%	<30%

Question 1

Which situation is an example of a force acting over a large area to produce a small pressure?

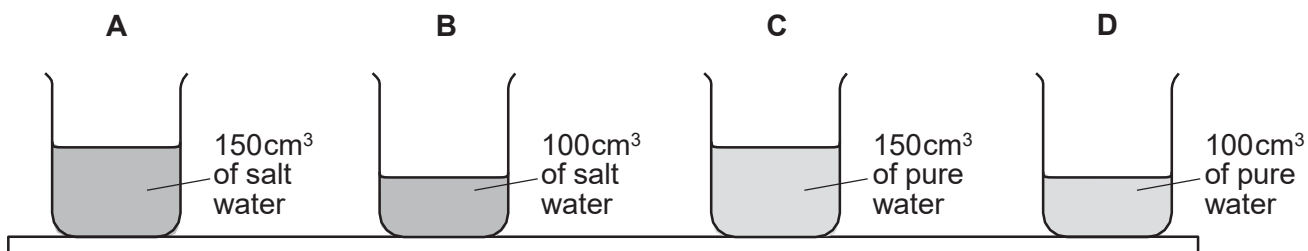
- A. a builder hammering a nail into a piece of wood
- B. a cook using a sharp knife to cut vegetables
- C. a nurse pushing a needle into a patient's arm
- D. a soldier marching in flat-soled boots

Question 2

A student places four identical beakers on a bench.

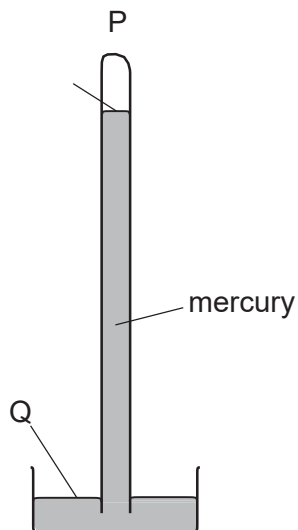
Two beakers contain salt water of density 1.1 g / cm^3 and two beakers contain pure water of density 1.0 g / cm^3 .

Which beaker exerts the greatest pressure on the bench?



Question 3

The diagram shows a simple mercury barometer.



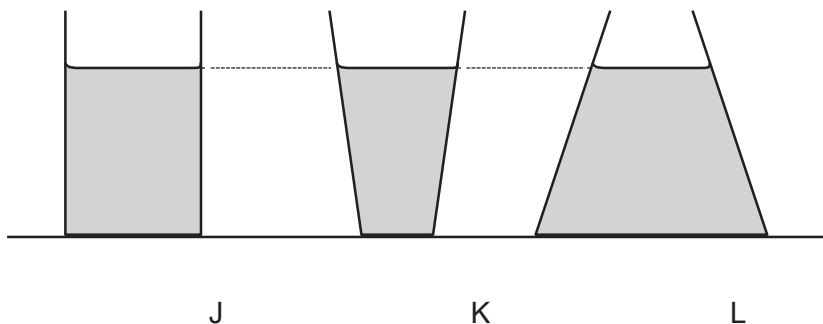
Atmospheric pressure decreases.

What happens to the level of the mercury at P and what happens to the level of the mercury at Q?

	P	Q
A	falls	falls
B	falls	rises
C	rises	falls
D	rises	rises

Question 4

The diagram shows three different containers J, K and L. Each container contains water of the same depth.

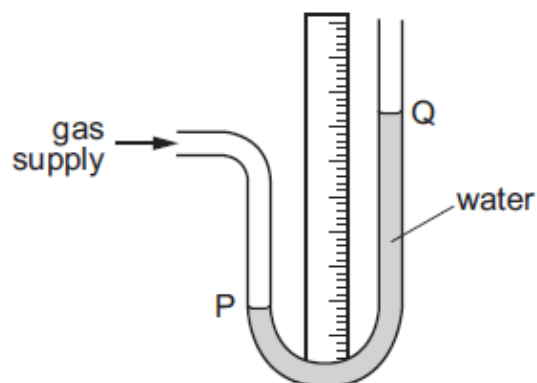


Which statement about the pressure of the water on the base of each container is correct?

- A The water pressure is greatest in container J.
- B The water pressure is greatest in container K.
- C The water pressure is greatest in container L.
- D The water pressure is the same for all three containers.

Question 5

A water manometer is connected to a gas supply.



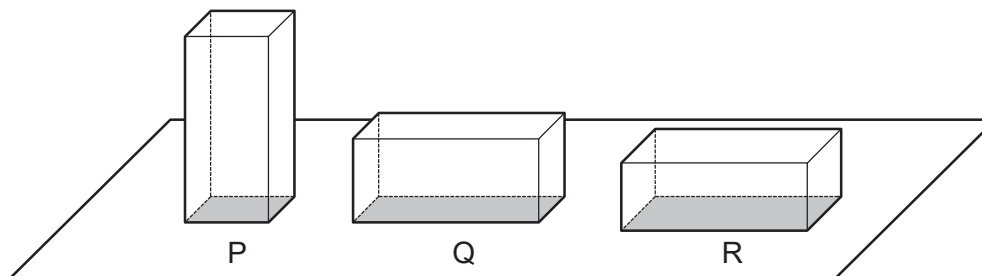
There is a gas leak and the pressure of the gas supply falls.

What happens to the water level at P and what happens to the water level at Q?

	water level at P	water level at Q
A	falls	falls
B	falls	rises
C	rises	falls
D	rises	rises

Question 6

The scale diagram shows three identical blocks, P, Q and R. The blocks have different areas of contact with the ground.

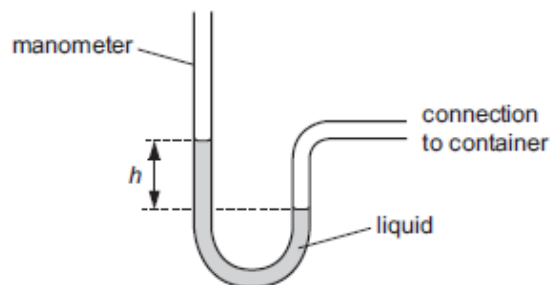


Which block exerts the greatest pressure on the ground?

- A. block P
- B. block Q
- C. block R
- D. they all exert the same pressure

Question 7

A manometer is used to measure the pressure of the air in a container.

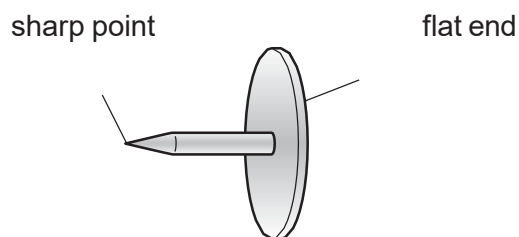


Which change would give a bigger value of height h ?

- A. using a less dense liquid
- B. using a more dense liquid
- C. using a narrower tube
- D. using a wider tube

Question 8

A drawing pin (thumb tack) has a sharp point and a flat end.



The pin is pushed into a wooden board.

How do the pressure and the force at the sharp point compare with the pressure and the force at the flat end?

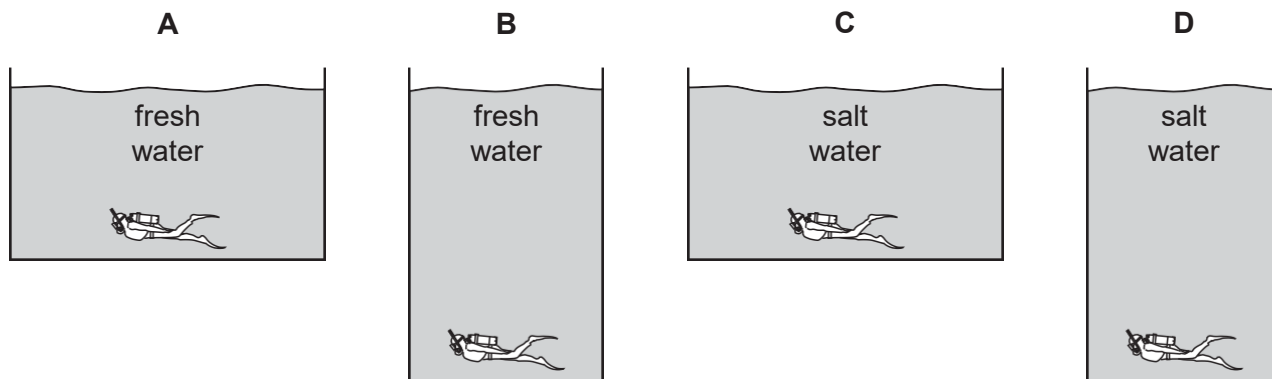
	force at the sharp point	pressure at the sharp point
A	greater than at the flat end	greater than at the flat end
B	greater than at the flat end	less than at the flat end
C	the same as at the flat end	greater than at the flat end
D	the same as at the flat end	less than at the flat end

Question 9

The diagrams show four divers at the bottom of four different swimming pools.

Two swimming pools contain fresh water and two contain salt water. Fresh water is less dense than salt water.

Which diver feels the least pressure from the water?



Question 10

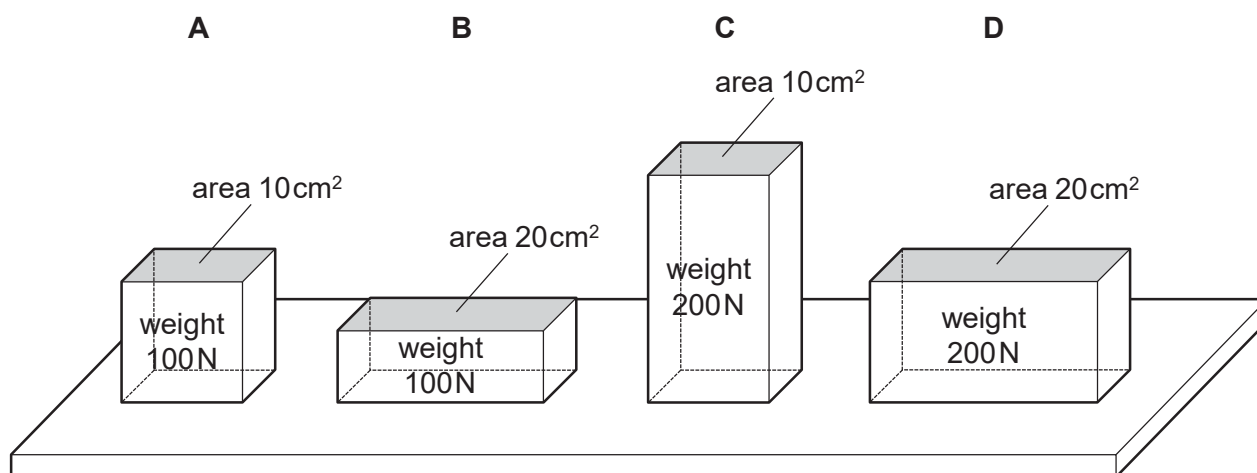
It is dangerous for submarines to dive to very great depths.

Why is it dangerous?

- A. The density of water is less at greater depths.
- B. The pressure of water is greater at greater depths.
- C. The temperature of water is higher at greater depths.
- D. The weight of the submarine is greater at greater depths.

Question 11

Which block exerts the greatest pressure on the surface below it?



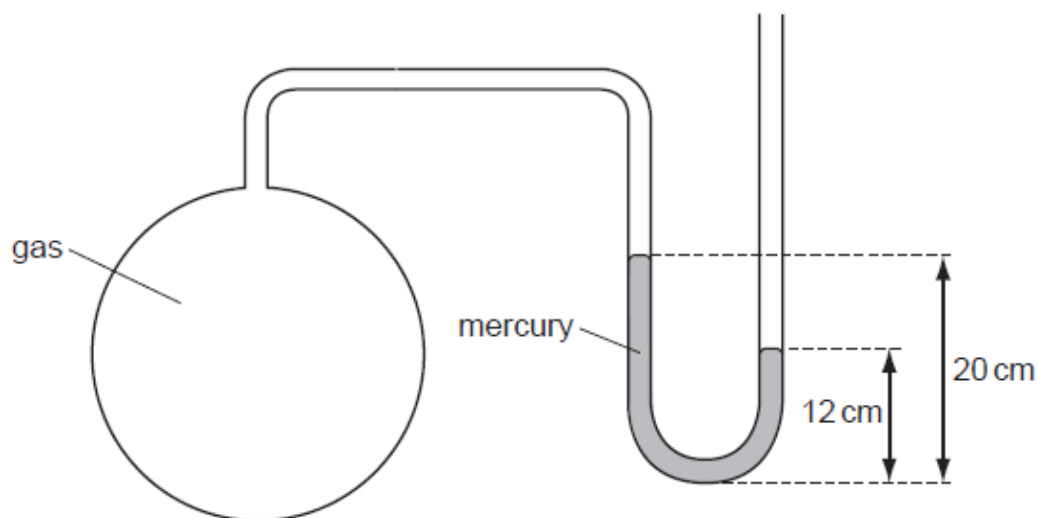
Question 12

Which statement is explained by reference to pressure?

- A. Objects with greater mass have greater weight.
- B. One kilogram of water occupies more volume than one kilogram of lead.
- C. Spikes on running-shoes sink into the ground.
- D. Water cooled to a low enough temperature turns to ice.

Question 13

The diagram shows a mercury manometer used to measure the pressure of gas in a container. Atmospheric pressure is 76 cm of mercury.



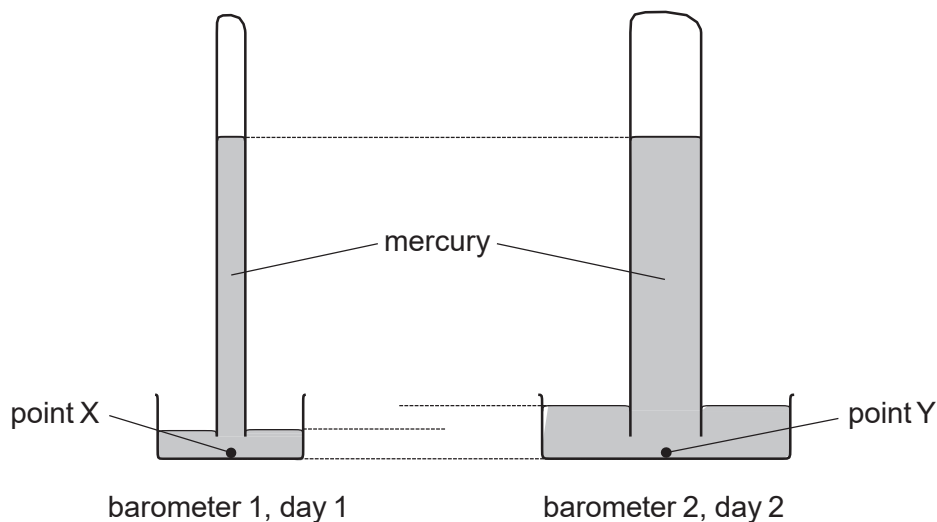
- A. 56 cm of mercury
- B. 68 cm of mercury
- C. 84 cm of mercury
- D. 96 cm of mercury

Question 14

The diagram shows two mercury barometers.

Barometer 1 is measuring atmospheric pressure on day 1.

Barometer 2 is measuring atmospheric pressure on day 2.

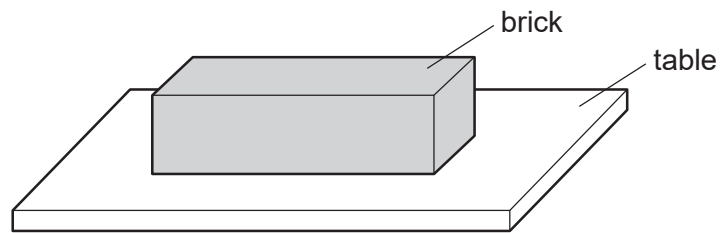


Which statement is true?

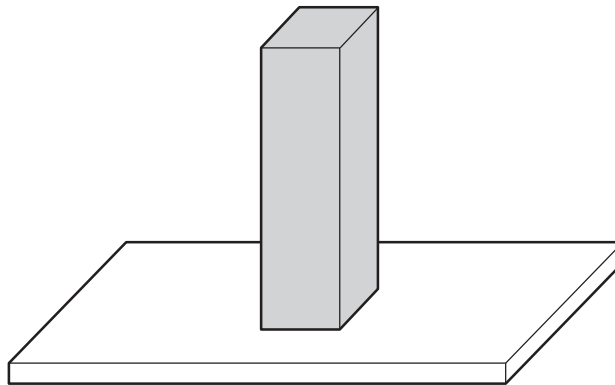
- A. The atmospheric pressure on day 1 is less than the atmospheric pressure on day 2.
- B. The atmospheric pressure on day 1 is the same as the atmospheric pressure on day 2.
- C. The pressure at point X is less than the pressure at point Y.
- D. The pressure at point X is the same as the pressure at point Y.

Question 15

A brick with flat, rectangular sides rests on a table.



The brick is now turned so that it rests on the table on its smallest face.

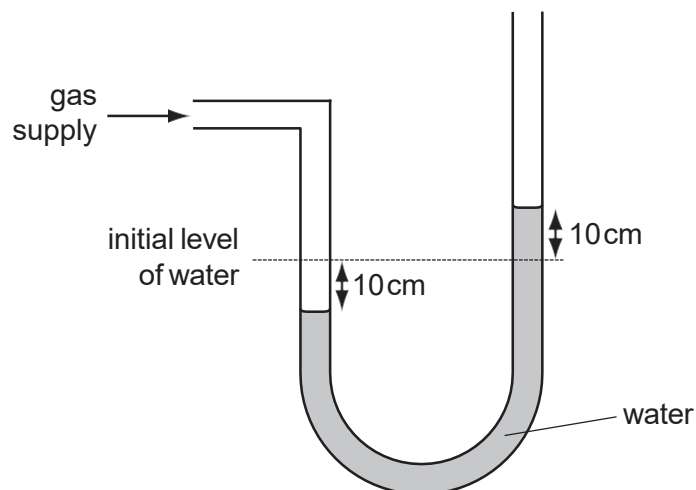


How has this affected the force and the pressure exerted by the brick on the table?

	force	pressure
A	increased	increased
B	increased	unchanged
C	unchanged	increased
D	unchanged	unchanged

Question 16

A water manometer is used to measure the pressure of a gas supply.



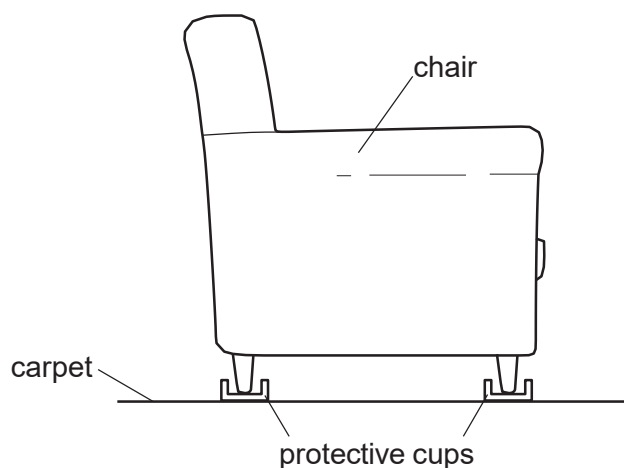
When it is attached to the gas supply, the water falls on the left side and rises on the right side. The difference in the levels of water on the two sides is now 20 cm.

What is the pressure of the gas supply?

- A. the pressure due to 10cm depth of water
- B. the pressure due to 20cm depth of water
- C. the pressure due to 10cm depth of water plus atmospheric pressure
- D. the pressure due to 20cm depth of water plus atmospheric pressure

Question 17

A chair is placed on protective cups to prevent damage to the carpet underneath it.



How do the cups change the area of contact with the carpet and the pressure on it?

	area of contact	pressure
A	decreased	decreased
B	decreased	increased
C	increased	decreased
D	increased	increased